Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

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Abstract

Kakutani's theorem tensorizes equivalence of product measures, but dense bounded linear mixing destroys the independence on which that test rests. The resulting difficulty is genuinely infinite-dimensional: every fixed finite output marginal may look regular while the full transformed law is singular. For a non-Gaussian i.i.d. product with positive density, finite second moment, and finite location and scale Fisher information, we prove that equivalence, Hilbert–Schmidt proximity to the discrete stabilizer $\mathcal{P}$, and bounded projected Hellinger energy are equivalent. The weighted Fourier jet supplies finite-frequency, dimension-free Hellinger witnesses that turn local mixing into Frobenius cost. We also classify standard symmetric α-stable products for $0 < \alpha < 2$. In this infinite-variance branch, the exact cost is anisotropic: diagonal deviations are square summable, while the ℓ^2-norms of the off-diagonal input columns are α-summable. Thus positivity alone does not make Hilbert–Schmidt proximity sufficient. When a finite-variance density has zeros, the Hilbert–Schmidt criterion has an exact support-aware extension: the rows of both T and T^{-1} must avoid the zero set. Half-line and bounded supports then reduce the equivalence group to explicit permutation classes.

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1 Introduction and Main Results

1.1 The question and answer

Fix one non-Gaussian coordinate law ρ and the i.i.d. product $\mu = \rho^{\otimes\mathbb{N}}$. For which bounded invertible linear maps T on ℓ^2 is the row-series image law ν_T equivalent to μ? Kakutani's product criterion answers coordinatewise perturbations, but it does not survive dense mixing: the transformed coordinates cease to be independent and the Hellinger affinity no longer tensorizes. The obstruction is not visible from any fixed finite window, since every fixed finite output marginal can have a strictly positive density even when the full transformed law is singular. Under the finite-variance and Fisher hypotheses of the main theorem, the continuous Gaussian symmetry collapses to the discrete stabilizer $\mathcal{P}$, and the surviving continuous thickness is exactly Hilbert–Schmidt. Equivalently, the projected Hellinger energies $E_n(T)$, which retain the whole directed family of output marginals, are bounded exactly on that equivalence branch. This conclusion is not a consequence of density positivity alone. For standard symmetric α-stable coordinates with $0 < \alpha < 2$, the density is everywhere positive but the exact criterion has a stronger anisotropic cost: diagonal errors are square summable and the off-diagonal input-column norms are α-summable. When a finite-variance density satisfies the Sobolev assumptions but has zeros, the Hilbert–Schmidt condition must be supplemented by a support gate: the rows of both T and T^{-1} must avoid the zero set of the one-dimensional density.

1.2 Historical context: from Kakutani to dense linear mixing

Kakutani's theorem turns equivalence of infinite products into a scalar summability problem. The test is the product of one-coordinate Hellinger affinities, positive exactly on the equivalence branch and zero exactly on the singular branch [10]. Sharp criteria for sequences of Euclidean and, more generally, Lie-group perturbations retain this product structure and identify Fisher information as the local Hellinger metric [4, 15].

The Gaussian law conceals the dense-mixing obstruction. For the standard Gaussian product, every orthogonal operator on ℓ^2 has a measurable row-series realization and preserves the product law. Translation and linear measure-class changes are governed by the Cameron–Martin and Feldman–Hájek theories, with explicit Gaussian density criteria developed by Shepp and nonlinear extensions by Ramer [2, 3, 5, 7, 12, 14]. A non-Gaussian product behaves differently: exact independence rigidifies the orthogonal group to coordinate permutations and to those coordinate signs already preserving the one-dimensional law. Non-Gaussian quasi-invariant measures on Hilbert and linear topological spaces have long been known [6, 16]; the problem here is the complete measure-class classification for dense linear mixing of one fixed i.i.d. product law.

1.3 The phase transition: continuous Gaussian symmetry and discrete rigidity

Let X have a centered probability law ρ with finite first moment and characteristic function $\phi(u) = \mathbb{E}e^{iuX}$. The Fourier statistic used below has a specific job: it turns the dependence created by a small non-Gaussian rotation into a bounded observable whose strength does not deteriorate with the ambient dimension. Define the rotational Fourier defect by

$$R(\xi, \eta) = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \tag{1}$$

It is the first variation of the expectation of a product of centered complex exponentials under a planar rotation. Its determinant representation uses the weighted Fourier jet, the explicit curve

$$u \longmapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2. \tag{2}$$

Up to sign, $R(\xi, \eta)$ is the determinant of the two vectors on this curve at ξ and η. Hence zero rotational Fourier defect means that the curve lies in one fixed complex line through the origin. The resulting global linear ODE has only Gaussian characteristic functions as solutions.

Under the finite-second-moment and Sobolev assumptions below, a finite set of frequencies detects every infinitesimal matrix direction for a non-Gaussian coordinate law. The corresponding Hellinger orbit is uniformly coercive in Frobenius norm near the exact linear stabilizer, independently of dimension. In that finite-variance regime, the surviving continuous thickness is exactly Hilbert–Schmidt: equivalence holds precisely for Hilbert–Schmidt perturbations of coordinate permutations whose signs preserve ρ. The symmetric stable regime has the same discrete stabilizer but a different, anisotropic accumulation law, as shown in Section 7.

1.4 Main theorem: equivalence, energy, and the stabilizer

The main finite-variance theorem has a simple nontechnical content. For coordinate laws satisfying (3), dense mixing preserves the product measure class exactly when it differs by Hilbert–Schmidt size from an exact coordinate symmetry. The finite-marginal energy $E_n(T)$ is not a heuristic proxy for this condition; it is an equivalent diagnostic. Positivity alone does not imply this Hilbert–Schmidt criterion: the standard symmetric α-stable laws have everywhere positive densities, but their infinite-variance classification requires the stronger anisotropic cost in (93).

We now state the assumptions used for the full arbitrary-mixing theorem. Suppose $\rho(dx) = f(x)\,dx$, put $g = \sqrt{f}$, and assume

$$\begin{gathered} f > 0 \quad \text{Lebesgue-almost everywhere}, \qquad g \in W^{1,2}(\mathbb{R}), \qquad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1, \qquad f \neq (2\pi)^{-1/2} e^{-x^2/2}. \end{gathered} \tag{3}$$

All derivatives are weak derivatives. Cusps are allowed. No moment of order greater than two is assumed. The two Sobolev conditions are finite location and scale Fisher information.

Put $H = \ell^2(\mathbb{N};\mathbb{R})$ and $\mu = \rho^{\otimes\mathbb{N}}$ on the product Borel space $\mathbb{R}^{\mathbb{N}}$, and $\mu_n = \rho^{\otimes n}$. The space H indexes coefficients. Write $\mathcal{B}(H)$ for its bounded linear operators, $\mathrm{GL}(H)$ for the invertible members, and $(e_i)_{i\geq 1}$ for the standard coordinate basis. A sample from μ is not asserted to belong to H. Define

$$\begin{aligned} \mathcal{P} = \{Q : (Qx)_i = \varepsilon_i x_{\pi(i)}, \quad &\pi \text{ a permutation of } \mathbb{N}, \\ \varepsilon_i \in \{-1, 1\}, \quad &\mathrm{Law}(\varepsilon_i X) = \rho \text{ for every } i\}. \end{aligned} \tag{4}$$

Thus $\mathcal{P}$ is the full signed-permutation group when ρ is symmetric, and the ordinary permutation group otherwise. This definition also applies when ρ has no density. For a bounded linear operator T on H, let Φ_T be the measurable row-series realization constructed in Theorem 4.1. Write $\nu_T = (\Phi_T)_{\#}\mu$.

For probability measures λ, ν, set

$$\mathsf{A}(\lambda,\nu) = \int \sqrt{\frac{d\lambda}{dm}\frac{d\nu}{dm}}\, dm, \qquad h(\lambda,\nu)^2 = 2 - 2\mathsf{A}(\lambda,\nu), \tag{5}$$

where m dominates both measures. We use the same notation for densities and their probability measures. Let $P_n : H \to H$ be the orthogonal projection onto the first n coordinate vectors, and let $\pi_n : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^n$ extract the first n coordinates. Write $\mathcal{S}_2(H)$ for the Hilbert–Schmidt ideal. Define the projected Hellinger energy by

$$E_n(T) = -2\log \mathsf{A}\big(\mu_n, (\pi_n)_{\#}\nu_T\big). \tag{6}$$

These are actual output marginals, not laws obtained from principal matrix corners. Thus E_n keeps the finite evidence produced by the true dense output rows and asks whether that evidence remains uniformly bounded as the window grows.

Theorem 1.1 (Complete criterion for arbitrary coordinate mixing)**.** *Under (3), for every $T \in \mathrm{GL}(H)$,*

$$\boxed{\begin{aligned} \nu_T \sim \mu \quad &\Longleftrightarrow \quad \exists Q \in \mathcal{P} :\; TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\Longleftrightarrow \quad \sup_n E_n(T) < \infty. \end{aligned}} \tag{7}$$

If these conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. The exact bounded invertible linear stabilizer is $\mathcal{P}$: $\nu_T = \mu$ if and only if $T \in \mathcal{P}$.

Proposition 1.2 (Finite-coordinate likelihood approximation)**.** *Under the assumptions of Theorem 1.1, suppose $\nu_T \sim \mu$. For any admissible Q, put $K = TQ^{-1} - I$ and $T_n = I + P_n K P_n$. Then T_n is eventually invertible and*

$$\frac{d\nu_{T_n}}{d\mu} \longrightarrow \frac{d\nu_{TQ^{-1}}}{d\mu} \quad \text{in } L^1(\mu), \qquad \sqrt{\frac{d\nu_{T_n}}{d\mu}} \longrightarrow \sqrt{\frac{d\nu_{TQ^{-1}}}{d\mu}} \quad \text{in } L^2(\mu). \tag{8}$$

Since $\nu_Q = \mu$, the limiting law is also ν_T. Moreover, the finite-marginal likelihood ratios and their reciprocals are uniformly integrable in their respective reference measures.

Theorem 1.2 is the analytic bookkeeping behind the criterion: finite-dimensional likelihood computations do not merely approximate cylinder probabilities, but converge to the actual infinite-dimensional likelihood on the equivalence branch.

The theorem contains the global energy formulation through $\sup_n E_n(T)$. The rotational Fourier defect is not an additional scalar condition on T. It is the calibration mechanism. It turns small Hellinger displacement into squared Frobenius displacement. This distinction keeps the one-dimensional Gaussian characterization separate from the operator-valued equivalence criterion.

Two refinements identify the assumptions needed for the stabilizer and for orthogonal transformations.

Corollary 1.3 (Moment-only stabilizer)**.** *If ρ is centered, variance one, and non-Gaussian, then the bounded invertible linear stabilizer of $\mu = \rho^{\otimes \mathbb{N}}$ is $\mathcal{P}$. No density or Fisher-information assumption is needed.*

Corollary 1.4 (Equivalence under orthogonal transformations)**.** *If $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and f is centered, variance one, and non-Gaussian, then every orthogonal U obeys*

$$\nu_U \sim \mu \quad \Longleftrightarrow \quad U - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}.$$

Thus scale Fisher information is needed for general linear mixing but not for the orthogonal subproblem.

1.5 Result map

The classification separates metric accumulation from support compatibility. The following map records the strongest conclusion in each regime and the reason that conclusion matters.

Condition	**Strongest conclusion**	**Value**
Finite-variance assumptions (3), arbitrary bounded invertible mixing	$\nu_T \sim \mu \iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2 \iff \sup_n E_n(T) < \infty$	A complete and checkable measure-class boundary for dense mixing.
Centered, variance-one, non-Gaussian positive density with finite location Fisher information; orthogonal mixing	The same Hilbert–Schmidt criterion, assuming only location Fisher information.	The important rotation regime needs weaker regularity.
Standard symmetric α-stable coordinates, $0 < \alpha < 2$; compatible row action and inverse	$\nu_T^{(\alpha)} \sim \mu_\alpha \iff T\|_H \in \mathrm{GL}(H)$, $\mathfrak{e}_\alpha(Q^{-1}T) < \infty$ for some $Q \in \mathcal{P} \iff \sup_n E_{\alpha,n}(T) < \infty$.	Diagonal errors have quadratic cost, while off-diagonal input columns have $\ell^\alpha(\ell^2)$ cost; Hilbert–Schmidt proximity alone is not sufficient.
Finite-variance Sobolev assumptions (74); density with zeros	Hilbert–Schmidt proximity plus row-wise avoidance of the zero set by T and T^{-1}.	Positive affinity is separated from true two-sided absolute continuity.
Half-line or bounded support	The equivalence group collapses to explicit permutation or signed-permutation classes.	Support boundaries can eliminate nontrivial dense mixing altogether.

The first row is the finite-variance main theorem. The second isolates the common situation in which the operator is already orthogonal, so the scale score never enters. The third is the infinite-variance stable classification: its columnwise cost records the heavy-tailed source coordinate shared by all receiving rows. The fourth is the support-aware replacement for positivity; it says exactly what null-set obstruction must be checked in addition to Hilbert–Schmidt size. The fifth shows that in one-sided and bounded geometries the support gate can be so rigid that the continuous Hilbert–Schmidt neighborhood disappears.

1.6 Organization and notation

Section 2 rules out continuous non-Gaussian rotation blind spots. Section 3 turns that Fourier witness into dimension-free local Hellinger coercivity and then states the finite-dimensional global form. Section 4 builds the measurable row-series action and shows that the projected Hellinger energy is an exact equivalence diagnostic. Section 5 proves sufficiency by constructing stable likelihood ratios and proves necessity by showing that non-Hilbert–Schmidt tail errors cannot be hidden from local coercivity and tail symmetry. Section 6 gives the support-aware replacement for density positivity, including the row zero-set criterion, boundary contrasts, and the bounded-support and half-line classifications. Section 7 then separates the remaining finite-variance assumptions and examples from the support geometry, and treats standard symmetric α-stable coordinates, where an anisotropic diagonal-square/off-diagonal-column cost replaces the Hilbert–Schmidt criterion.

We write A^* for the adjoint, including the transpose of a real matrix, and $\|\cdot\|_{\mathrm{op}}$ for the operator norm. The Frobenius inner product is $\langle A, B\rangle_F = \mathrm{tr}(A^*B)$, with norm $\|A\|_F$. The law

ρ remains fixed unless a different law is explicitly specified.

Symbol	Meaning
$\rho,\ \mu,\ \mu_n$	coordinate law and its infinite and n-coordinate products
ϕ	characteristic function $u \mapsto \mathbb{E}e^{iuX}$
R	rotational Fourier defect in (1)
$\mathsf{A},\ h$	Hellinger affinity and Hellinger distance
$P_n,\ \pi_n$	orthogonal projection on H and coordinate extraction on $\mathbb{R}^{\mathbb{N}}$
$E_n(T)$	projected Hellinger energy in (6)
$\mathcal{S}_2$	Hilbert–Schmidt ideal; $\|\cdot\|_{\mathcal{S}_2} = \|\cdot\|_F$ in finite dimensions
$\mathcal{P}$	coordinate permutations with signs preserving ρ
Φ_T, ν_T	measurable row-series action and its pushforward of μ
$\rho_\alpha,\ \mu_\alpha$	standard symmetric α-stable coordinate law and its product
$\mathfrak{c}_\alpha$	stable diagonal-square and off-diagonal-column mixing cost
$\Phi_T^{(\alpha)},\ \nu_T^{(\alpha)},\ E_{\alpha,n}$	stable row action, image law, and projected Hellinger energy

2 Rotational Fourier Defect and Gaussian Characterization

This section rules out the only way a non-Gaussian law could imitate the Gaussian rotation symmetry at first order. If every infinitesimal two-coordinate rotation were invisible to bounded Fourier statistics, the coordinate law would have to be Gaussian. This is the identification input later used by the dimension-free coercivity argument. The product of centered complex exponentials defining this defect is bounded. Its definition uses no logarithm of the characteristic function and remains valid when the characteristic function has zeros. The conclusion belongs to the same broad lineage as the Kac–Bernstein theorem and the classical characterization theory of independent linear forms [8, 9]. Its hypothesis is different. The rotational Fourier defect vanishes at every frequency pair. This leads directly to a global linear ODE.

2.1 A product of centered complex exponentials

This subsection requires only $\mathbb{E}X = 0$ and $\mathbb{E}|X| < \infty$. Neither a density nor a second moment is needed. Write $\phi(u) = \mathbb{E}e^{iuX}$, and for independent copies X_1, X_2 put

$$O_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad W_{\xi,\eta}(x,y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

This fixes the sign convention for rotation. Centering removes the two marginal responses. It leaves only the dependence created by rotation. In particular,

$$\mathbb{E}W_{\xi,\eta}(X_1,X_2) = 0, \qquad \mathbb{E}|W_{\xi,\eta}(X_1,X_2)|^2 = (1-|\phi(\xi)|^2)(1-|\phi(\eta)|^2) \le 1. \tag{9}$$

Pointwise, $|W_{\xi,\eta}| \le 4$, so its variance under an arbitrary comparison law is at most 16. Both bounds are dimension-free and require no tail assumption.

2.2 Gaussian characterization by a global linear ODE

The following lemma is the point at which the Fourier statistic becomes a rigidity tool. Later arguments use its contrapositive: if the coordinate law is not Gaussian, then some rotational direction has a nonzero first-order response. The formulation stays at the level of characteristic functions, so it avoids choosing a branch of $\log \phi$ and remains valid even when ϕ has zeros.

Lemma 2.1 (Gaussian characterization by the rotational Fourier defect)**.** *For the product of centered complex exponentials defined above,*

$$R(\xi, \eta) = \frac{d}{d\theta} \mathbb{E} W_{\xi,\eta}(O_\theta(X_1, X_2))\Big|_{\theta=0} = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \tag{10}$$

Moreover,

$$R \equiv 0 \quad \Longleftrightarrow \quad \mathrm{Law}(X) \textit{ is a centered Gaussian, possibly degenerate.} \tag{11}$$

Proof. The first moment implies $\phi \in C^1(\mathbb{R})$. The joint characteristic function after rotation is

$$\phi(\xi \cos\theta + \eta \sin\theta)\phi(-\xi \sin\theta + \eta \cos\theta).$$

The marginal characteristic functions are $\phi(\xi \cos\theta)\phi(-\xi \sin\theta)$ and $\phi(\eta \sin\theta)\phi(\eta \cos\theta)$. Their derivatives at zero vanish because $\phi'(0) = i\mathbb{E}X = 0$. Differentiating the joint characteristic function therefore gives (10).

If R vanishes everywhere, continuity and $\phi(0) = 1$ allow us to choose $\eta_0 \neq 0$ with $\phi(\eta_0) \neq 0$. Fix this single frequency. The defect equation then becomes the global ODE

$$\phi'(\xi) = c\xi\phi(\xi), \qquad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \qquad \phi(0) = 1.$$

No division by $\phi(\xi)$ has occurred. Uniqueness for this linear ODE gives $\phi(\xi) = \exp(c\xi^2/2)$ on all of $\mathbb{R}$, including any putative zero of ϕ. The identity $\phi(-\xi) = \overline{\phi(\xi)}$ forces $c \in \mathbb{R}$. Also $|\phi(\xi)| \leq 1$ forces $c \leq 0$. Write $c = -\sigma^2$. The Gaussian conclusion follows. If $\sigma = 0$, we get the point mass at zero. Substitution proves the converse. □

2.3 The determinant representation and frequency integrals

This reformulation is included because the finite-dimensional proof needs a finite set of test frequencies, not merely an abstract nonvanishing defect. The determinant viewpoint identifies the rank-two obstruction used to build the dual moments in Section 3.3.

Consider the curve

$$u \longmapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2,$$

described as the weighted Fourier jet. Its values satisfy

$$\det \begin{pmatrix} \xi\phi(\xi) & \phi'(\xi) \\ \eta\phi(\eta) & \phi'(\eta) \end{pmatrix} = -R(\xi, \eta). \tag{12}$$

Thus the rotational Fourier defect is the negative determinant of two values of this curve. In the Gaussian case $\phi'(u) = -\sigma^2 u\phi(u)$. So the curve lies in the fixed line spanned by $(1, -\sigma^2)$. Conversely, vanishing of every such determinant forces precisely the ODE used above.

The weighting by u is essential: for a nondegenerate Gaussian law, the unweighted curve $u \mapsto (\phi(u), \phi'(u))$ is not contained in a fixed line.

For a non-Gaussian law, some two distinct frequencies have nonzero defect. By continuity, a compact neighborhood of this pair already detects the law. Thus a compact frequency set may be fixed after the coordinate law is chosen. This set is independent of the ambient dimension. The diagonal $\xi = \eta$ always has zero defect and cannot serve this purpose. One may also integrate the squared defect over all frequencies:

$$\iint |R(\xi,\eta)|^2 w(\xi,\eta)\, d\xi\, d\eta, \qquad w > 0 \text{ a.e.}, \quad \iint (1+\xi^2+\eta^2) w(\xi,\eta)\, d\xi\, d\eta < \infty. \tag{13}$$

Indeed, $|R(\xi,\eta)| \le \mathbb{E}|X|(|\xi| + |\eta|)$. Hence this integral is finite and is zero exactly for centered Gaussian laws. Integrability of w alone need not make the integral finite.

The same product of centered complex exponentials gives a local Hellinger lower bound for one rotation. Choose a phase α with $\mathrm{Re}(e^{-i\alpha} R(\xi,\eta)) > 0$ and apply

$$|\mathbb{E}_\lambda F - \mathbb{E}_\nu F| \le h(\lambda,\nu)\{2\mathbb{E}_\lambda F^2 + 2\mathbb{E}_\nu F^2\}^{1/2}$$

to $F = \mathrm{Re}(e^{-i\alpha} W_{\xi,\eta})$. It follows that $h((O_\theta)_\# \operatorname{Law}(X)^{\otimes 2}, \operatorname{Law}(X)^{\otimes 2}) \ge c|\theta|$ for small $|\theta|$. A first moment suffices for this lower bound. The dimension-free matrix estimates below require the additional second-moment and Sobolev hypotheses.

2.4 Dimension-free variance and affine normalization

Equation (9) is the variance estimate behind the finite-frequency tests used below. Averaging N independent pairs gives standard error at most $N^{-1/2}$ under the reference product. The rough bound $16/N$ remains available under an arbitrary comparison distribution. In particular, the product of centered complex exponentials does not require a fourth moment, symmetry, or a density.

Centering, scaling, and phase. Let Y have mean m and characteristic function ϕ_Y, and put $X = (Y - m)/\sigma$, where $\sigma > 0$. For $a = \xi/\sigma$ and $b = \eta/\sigma$, the rotational Fourier defect of X satisfies

$$\begin{aligned} R(\xi,\eta) = e^{-i(m/\sigma)(\xi+\eta)} \big[& b\phi_Y'(a)\phi_Y(b) - a\phi_Y(a)\phi_Y'(b) \\ & + im(a-b)\phi_Y(a)\phi_Y(b)\big]. \end{aligned} \tag{14}$$

The expression in brackets is the derivative at zero of the expectation of the product of centered complex exponentials after rotating two independent copies of Y. Its last term comes from differentiating the marginal terms. Thus centering rotates this mean-corrected expression by $-(m/\sigma)(\xi+\eta)$. Scaling also changes the frequencies. At fixed corresponding frequencies its modulus is unchanged. Omitting the marginal correction would invalidate this phase identity. For a centered symmetric law the defect is real. At a particular frequency pair, an asymmetric law can still have a real or zero defect. So the phase is not, by itself, a complete measure of skewness.

For the concrete comparison $Y \sim \mathrm{Gamma}(2,1)$, $m = 2$, $\sigma = \sqrt{2}$, and $(\xi,\eta) = (1,2)$, the expression in brackets in (14) equals $4/27$, so the standardized law has

$$R(1,2) = \frac{4}{27} e^{-3i\sqrt{2}}.$$

At this particular pair, the entire phase of the rotational Fourier defect is therefore the centering phase, namely $-3\sqrt{2}$ modulo 2π. The variance-one Laplace law has rotational Fourier defect $R(1,2) = -4/27$. Their moduli agree at these frequencies, while their principal phases are approximately 116.915° and 180°, respectively.

3 Finite-Dimensional Hellinger Coercivity

We now convert the rotational Fourier defect into a matrix norm bound. The argument has two layers. The matrix Fisher operator identifies the infinitesimal geometry. Finite real Fourier sums provide coordinate functions whose dual moments give dimension-free bounds. The resulting local estimate is then globalized modulo the discrete stabilizer in Theorem 3.6.

3.1 The matrix Fisher operator

The Fisher operator supplies the infinitesimal metric that all later Hellinger bounds linearize around. Its eigenspaces separate diagonal scale directions, symmetric off-diagonal covariance directions, and skew-symmetric rotational directions. The rotational eigenvalue is $J-1$, so Gaussianity is exactly the case in which rotations lose their local cost.

Let f be a probability density and write $g = \sqrt{f}$. In this subsection we assume

$$\begin{gathered} f > 0 \quad \text{Lebesgue-almost everywhere}, \qquad g \in W^{1,2}(\mathbb{R}), \qquad xg'(x) \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1, \qquad X \sim f(x)\,dx. \end{gathered} \tag{15}$$

The derivatives in these assumptions are weak derivatives. Define the location score $s = -f'/f = -2g'/g$ almost everywhere. Then

$$\mathbb{E}s(X) = 0, \qquad \mathbb{E}[X s(X)] = 1, \qquad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2. \tag{16}$$

Define the scale score and its second moment by

$$d(x) := xs(x) - 1, \qquad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \tag{17}$$

The two integration-by-parts identities in (16) follow from these assumptions. Indeed, $f' = 2gg'$ belongs to L^1, and xf' belongs to L^1 by Cauchy–Schwarz and the finite second moment. Choose a smooth compactly supported cutoff χ equal to one near zero. Applying integration by parts first to $\chi(x/L)$ and then to $x\chi(x/L)$ gives the identities as $L \to \infty$.

For independent copies $X_1, \dots, X_n$ of X, write $\mathbf{X} = (X_1, \dots, X_n) \sim \mu_n$. For $A \in M_n(\mathbb{R})$, define the matrix score as the score of the pushforward by $I + tA$ at $t = 0$ is

$$S_A(\mathbf{X}) = \sum_{i,j=1}^{n} A_{ij}\big(s(X_i)X_j - \delta_{ij}\big) = \sum_i A_{ii} d(X_i) + \sum_{i \neq j} A_{ij} s(X_i) X_j. \tag{18}$$

Theorem 3.1 (The matrix Fisher operator)**.** *Under (15), define the matrix Fisher operator $\mathcal{I}$ by $\langle A, \mathcal{I}B\rangle_F = \mathbb{E}[S_A S_B]$. Then*

$$\boxed{\mathcal{I}A = JA + A^* + (\kappa - J - 1)\operatorname{diag} A.} \tag{19}$$

Here $\operatorname{diag} A$ *is the matrix with diagonal entries* A_{ii} *and zero off-diagonal entries. The diagonal, symmetric off-diagonal, and skew-symmetric matrix subspaces are orthogonal for the Frobenius inner product and have eigenvalues* κ, $J+1$, *and* $J-1$, *respectively. Equivalently,*

$$\begin{aligned} \langle A, \mathcal{I}A \rangle_F = \kappa \| \operatorname{diag} A\|_F^2 + (J+1) \left\| \frac{A+A^*}{2} - \operatorname{diag} A \right\|_F^2 \\ + (J-1) \left\| \frac{A-A^*}{2} \right\|_F^2 . \end{aligned} \tag{20}$$

The eigenvalue multiplicities are, respectively,

$$n, \qquad n(n-1)/2, \qquad n(n-1)/2. \tag{21}$$

Moreover, $\kappa > 0$ *and* $J \geq 1$*. Here* $J = 1$ *holds precisely for the standard Gaussian density. Thus, for a non-Gaussian* f, $\min\{\kappa, J-1\} > 0$ *and*

$$\begin{aligned} \min\{\kappa, J-1\} \|A\|_F^2 \leq \langle A, \mathcal{I}A \rangle_F \leq C_f \|A\|_F^2, \\ C_f := \max\{\kappa, J+1\}. \end{aligned} \tag{22}$$

Proof. The diagonal variables $d(X_i)$ are independent and centered. Their covariances with every off-diagonal term $s(X_j)X_k$, $j \neq k$, vanish. After any shared coordinate is integrated out, the remaining factor has expectation $\mathbb{E}X$ or $\mathbb{E}s(X)$. Two off-diagonal terms can have a nonzero covariance only when their ordered index pairs coincide or are reversed. Those covariances are, respectively, J and $(\mathbb{E}Xs(X))^2 = 1$. Consequently

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i<j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \tag{23}$$

Resolving each pair into its symmetric and skew-symmetric parts proves the decomposition. In particular, the calculation uses centering and independence and imposes no parity condition on f.

Cauchy–Schwarz applied to $\mathbb{E}Xs(X) = 1$ gives $J \geq 1$. Equality implies $s(X) = X$ almost surely, hence $g' = -xg/2$ almost everywhere and f is the standard Gaussian. If $\kappa = 0$, then $2xg' + g = 0$ almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of $|x|^{-1/2}$, which cannot give a positive probability density. This proves $\kappa > 0$ and the asserted bounds. □

3.2 Hellinger paths, polar decomposition, and symmetric perturbations

For orientation, write the left polar decomposition of an invertible matrix as $T = BO$, where $B = (TT^*)^{1/2}$ is positive and O is orthogonal. The covariance defect is $TT^* - I = B^2 - I$, so a finite second moment detects the symmetric factor. The skew tangent of the orthogonal factor has Fisher eigenvalue $J-1$, which is positive precisely off the Gaussian law. The full matrix Fisher decomposition above packages these two mechanisms without requiring a nonlinear separation of B and O at every step.

For $A \in \mathrm{GL}(n, \mathbb{R})$ define a unitary operator on $L^2(\mathbb{R}^n, dx)$ by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \qquad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$

Thus $U_A\Omega_n$ is the square root of the Lebesgue density of $A_{\#}\mu_n$. For invertible A, B, $h(A_{\#}\mu_n, B_{\#}\mu_n) = \|U_A\Omega_n - U_B\Omega_n\|_2$. For a fixed real matrix A, the generator of $U_{\exp(tA)}$ is

$$G_A v = -\tfrac{1}{2}\operatorname{tr}(A)v - (Ax)\cdot\nabla v, \qquad G_A\Omega_n = \tfrac{1}{2}S_A\Omega_n. \tag{24}$$

These identities hold in L^2. To justify them under (15), observe that $x_j\partial_i\Omega_n \in L^2$ for all i, j: the diagonal case uses $xg' \in L^2$, and the off-diagonal case uses $g' \in L^2$ and $\mathbb{E}X^2 = 1$. Cutoff and mollification approximate Ω_n in the corresponding weighted Sobolev graph norm, where the usual change-of-variables differentiation is valid.

Lemma 3.2 (A dimension-free Hellinger path bound)**.** *If A_t, $0 \le t \le 1$, is a continuously differentiable path in $\mathrm{GL}(n,\mathbb{R})$, then $t \mapsto U_{A_t}\Omega_n$ is continuously differentiable in L^2 and*

$$\begin{aligned} \frac{d}{dt}U_{A_t}\Omega_n &= U_{A_t}G_{A_t^{-1}\dot{A}_t}\Omega_n, \\ \left\|\frac{d}{dt}U_{A_t}\Omega_n\right\|_2 &= \frac{1}{2}\sqrt{\left\langle A_t^{-1}\dot{A}_t, \mathcal{I}(A_t^{-1}\dot{A}_t)\right\rangle_F}. \end{aligned} \tag{25}$$

Consequently,

$$h((A_1)_{\#}\mu_n, (A_0)_{\#}\mu_n) \le \frac{\sqrt{C_f}}{2}\int_0^1 \|A_t^{-1}\dot{A}_t\|_F\, dt. \tag{26}$$

In particular, if $\|B\|_{\mathrm{op}} \le r < 1$, then

$$h((I+B)_{\#}\mu_n, \mu_n) \le \frac{\sqrt{C_f}}{2(1-r)}\|B\|_F. \tag{27}$$

All constants are independent of n.

Proof. The representation law $U_AU_B = U_{AB}$ and (24) give (25). Integrate the derivative and use the Fisher upper bound. For the last assertion take $A_t = I + tB$ and use $\|A_t^{-1}\|_{\mathrm{op}} \le (1-r)^{-1}$. $\square$

3.3 Finite real Fourier sums with dual moments

We now assume that f is non-Gaussian in addition to (15). We use finite real Fourier sums. These are finite real linear combinations of 1, $\cos(\xi x)$ and $\sin(\xi x)$ with real frequencies ξ. Their frequencies may depend on f. Once chosen, their finite set is independent of the matrix dimension.

The Fisher operator gives the correct quadratic form, but the lower bound must be witnessed by bounded coordinate functions whose variance does not grow with the dimension. The next lemma builds such witnesses from finitely many Fourier modes. The non-Gaussian Fourier defect from Theorem 2.1 is used exactly to make the off-diagonal moment map rank two.

Lemma 3.3 (Finite real Fourier sums with dual moments)**.** *There exist centered finite real Fourier sums p, q, r such that*

$$\begin{array}{c|cc} & s & x \\ \hline p & 1 & 0 \\ q & 0 & 1 \end{array} \qquad \textit{and} \qquad \mathbb{E}[r(X)d(X)] = 1. \tag{28}$$

The entries in the table denote expectations of the products. For

$$H_A(\mathbf{X}) := \sum_i A_{ii} r(X_i) + \sum_{i \neq j} A_{ij} p(X_i) q(X_j), \tag{29}$$

one has, for all real matrices A, B of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \qquad \|H_A\|_{L^2(\mu_n)} \leq C_0 \|A\|_F, \tag{30}$$

where C_0 is independent of n.

Proof. On the real vector space of centered finite real Fourier sums consider the two functionals

$$L_1(h) = \mathbb{E}h'(X) = \mathbb{E}[h(X)s(X)], \qquad L_2(h) = \mathbb{E}[Xh(X)].$$

The integration-by-parts identity holds since h, h' are bounded. Both functionals are unchanged by subtracting a constant. If this moment map had rank at most one, all its two-by-two determinants would vanish, including after complexification. Writing $\phi(\xi) = \mathbb{E}e^{i\xi X}$ and testing with the centered exponentials at frequencies ξ, η would then give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \qquad (\xi, \eta \in \mathbb{R}).$$

Here $L_1(e^{i\xi x}) = i\xi\phi(\xi)$ and $L_2(e^{i\xi x}) = -i\phi'(\xi)$. The global linear ODE argument in Theorem 2.1 would force a centered Gaussian law, a contradiction. The map therefore has rank two. Inverting the moment matrix of two finite real Fourier sums supplies p, q with the displayed dual moments. This construction fixes finitely many frequencies depending only on f.

The diagonal function can be chosen using only the second moment. For $\xi \neq 0$ let $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$. Dominated convergence gives

$$\frac{\mathbb{E}[Xr_0'(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \longrightarrow -\mathbb{E}X^2 = -1 \qquad (\xi \to 0).$$

Choose a sufficiently small nonzero ξ and set $r = r_0/\mathbb{E}[Xr_0'(X)]$. Cutoff integration by parts gives $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$.

All diagonal/off-diagonal cross terms in $\mathbb{E}H_A S_B$ vanish by centering. Coincident ordered off-diagonal pairs contribute $\mathbb{E}[ps]\mathbb{E}[qX] = 1$. Reversed pairs contribute $\mathbb{E}[pX]\mathbb{E}[qs] = 0$. This proves the pairing. For any centered $a, b \in L^2(f)$, independence gives

$$\left\| \sum_{i \neq j} A_{ij} a(X_i) b(X_j) \right\|_2 \leq \sqrt{2}\, \|a\|_2 \|b\|_2 \|A\|_F. \tag{31}$$

Together with the independent diagonal terms, this proves the last assertion. □

The functions H_A will be paired with the Hellinger tangent along a matrix path. To make that pairing stable after moving away from the identity, we need one uniform derivative bound for $H_A\Omega_n$. The next lemma is the technical price of keeping the test functions finite-frequency and dimension-free. Its proof gives the index contractions explicitly and uses no fourth moment of X or higher moment of the score.

Lemma 3.4 (Generator bound for the functions H_A)**.** *Let p, q, r be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define H_A by (29). There is a constant C_1, depending on these functions and on J, κ, such that*

$$\|G_B(H_A\Omega_n)\|_{L^2(dx)} \le C_1\|A\|_F\|B\|_F \qquad (A, B \in M_n(\mathbb{R}),\ n \ge 1). \tag{32}$$

Proof. For fixed n, both H_A and its first derivatives are bounded. Consequently

$$x_j\partial_i(H_A\Omega_n) = H_A x_j\partial_i\Omega_n + x_j(\partial_i H_A)\Omega_n \in L^2(dx)$$

by the weighted Sobolev regularity of Ω_n and $\mathbb{E}X^2 = 1$. The cutoff and mollification argument used for (24) therefore puts $H_A\Omega_n$ in the domain of G_B. The product rule gives

$$G_B(H_A\Omega_n) = \left\{\tfrac{1}{2}H_A S_B - (Bx)\cdot\nabla H_A\right\}\Omega_n. \tag{33}$$

Both terms belong to L^2 in each finite dimension. We prove a dimension-free bound for each term.

We will repeatedly use the following elementary consequence of independence. If $h_1, \dots, h_m$ are centered $L^2(f)$ functions, and a coefficient tensor C is indexed by ordered tuples of *distinct* coordinates, then

$$\left\|\sum_{i_1,\dots,i_m}^{\neq} C_{i_1\dots i_m}\prod_{a=1}^{m} h_a(X_{i_a})\right\|_2 \le \sqrt{m!}\,\|C\|_{\ell^2}\prod_{a=1}^{m}\|h_a\|_2. \tag{34}$$

In the square of this norm, a mixed term vanishes unless the two tuples have the same index set. There are at most $m!$ matching permutations. Cauchy–Schwarz for the local moments gives a bound on each index pattern. A second Cauchy–Schwarz step for the coefficient arrays proves the full bound. We use it only for $m \le 4$.

First consider $H_A S_B$. For a vector a and a zero-diagonal matrix M, abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \qquad Q_M(u, v) = \sum_{i\neq j} M_{ij}u(X_i)v(X_j).$$

The product $H_A S_B$ is the sum of the four products obtained by multiplying $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$ and $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$, where $\mathbf{a} = (A_{ii})_i$, $\mathbf{b} = (B_{ii})_i$, $A_0 = A - \operatorname{diag} A$, and $B_0 = B - \operatorname{diag} B$. In every common coordinate, a local product is a bounded function times one of d, s, x and hence is in $L^2(f)$. Write each such local product as its mean plus its centered part.

For a product $D_a(u)D_b(v)$, distinct indices give a coefficient tensor bounded by $\|a\|_2\|b\|_2$; equal indices give the Hadamard vector $(a_ib_i)_i$ and the scalar $\sum_i a_ib_i$. Their norms have the same bound. For $D_a(u)Q_M(v, w)$, there are three index patterns: three distinct indices, equality with the first index of Q_M, and equality with its second index. The first gives the tensor $a \otimes M$. In either equality case, centering the local product gives a matrix obtained by row or column multiplication of M by a, and its mean gives M^*a or Ma. Each norm is at most $\|a\|_2\|M\|_F$. The product $Q_M(u, v)D_a(w)$ has exactly the same three patterns and bounds.

For clarity, all patterns in a quadratic–quadratic product can also be enumerated. Consider

$$Q_M(u, v)Q_N(w, z) = \sum_{i\neq j,\, k\neq l} M_{ij}N_{kl}u_iv_jw_kz_l,$$

where u, v are bounded and w, z are in L^2. Four distinct indices give the restricted tensor $M \otimes N$, of norm at most $\|M\|_F \|N\|_F$. There are four patterns with exactly one shared index: $i = k$, $i = l$, $j = k$, and $j = l$. For example, the pattern $i = k$ is

$$\sum_{i,j,l}^{\neq} M_{ij} N_{il} (uw)_i v_j z_l.$$

Its centered local product has coefficient norm bounded by

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its local mean has coefficient matrix $(M^* N)_{jl}$ off the diagonal; excluding $i = j, l$ does not change the sum because M, N have zero diagonal. The other three shared-index patterns give the same estimate with a transpose or with the two coordinates exchanged, and their mean contractions are matrix products of M, N or their transposes.

Finally, there are two patterns with both indices shared: $(k, l) = (i, j)$ and $(k, l) = (j, i)$. Their coefficients are $M_{ij} N_{ij}$ and $M_{ij} N_{ji}$. After centering the two local products, the resulting coefficients are a Hadamard matrix, its row sums, its column sums, and its total sum. For instance

$$\sum_i \left| \sum_j M_{ij} N_{ij} \right|^2 \leq \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_j N_{ij}^2 \right) \leq \|M\|_F^2 \|N\|_F^2,$$

and the other three estimates follow from the same Cauchy–Schwarz inequality. These cases exhaust the possible index coincidences. Applying (34) to the centered pieces therefore proves

$$\|H_A S_B\|_2 \leq C \|A\|_F \|B\|_F. \tag{35}$$

It remains to bound $(Bx) \cdot \nabla H_A$. We give the contractions separately, since derivatives of a centered function need not be centered. For the diagonal part the sum is

$$\sum_{i,k} A_{ii} B_{ik} r'(X_i) X_k.$$

When $i \neq k$, split r' into its mean and centered part. The latter has coefficient matrix $(\operatorname{diag} A)B$, with its diagonal deleted. The mean has coefficient vector $B^* \mathbf{a} - \mathbf{a} \odot \mathbf{b}$. Their norms are bounded by, respectively, one and two times $\|A\|_F \|B\|_F$. When $i = k$, center the local function $x r'(x)$, which is in L^2 since r' is bounded. The coefficient vector and the mean are bounded by the same product of matrix norms.

Let $M = A_0$. The derivative of the off-diagonal part is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and $F(M^*, B; q', p)$. In F, first take k distinct from i, j and split p' into its mean and centered part. The centered part has three-index coefficient norm bounded by

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F.$$

The mean part has, off the diagonal, the matrix $M^*B - M^* \operatorname{diag} B$; the second term removes $i = k$, and $i = j$ already contributes zero. Its Frobenius norm is at most $2\|M\|_F\|B\|_F$. When $k = i$, center $xp'(x)$; the resulting matrix has entries $B_{ii}M_{ij}$ and its mean has vector $M^*\mathbf{b}$. When $k = j$, split both p' and $xq(x)$ into means and centered parts; the coefficients are the Hadamard matrix $M \odot B$, its row and column sums, and its total sum. The preceding Hadamard estimates bound all of them. All the local functions used here belong to L^2, since p', q are bounded and $\mathbb{E}X^2 = 1$. The same proof applies to $F(M^*, B; q', p)$. Using (34) again proves

$$\|(Bx) \cdot \nabla H_A\|_2 \le C\|A\|_F\|B\|_F.$$

Together with (35) and (33), this proves the lemma. □

We can now close the local argument. The dual Fourier functions give a linear functional that sees the first-order displacement, while the generator bound controls the error made along the exponential path. This turns the nonzero Fourier defect into a uniform Hellinger lower bound for every small matrix perturbation.

Proposition 3.5 (Uniform local Hellinger coercivity)**.** *For every non-Gaussian density satisfying* (15)*, there are constants $c, C, \delta > 0$, independent of n, such that*

$$c\|B\|_F \le h((I+B)_{\#}\mu_n, \mu_n) \le C\|B\|_F \qquad \text{whenever } \|B\|_F \le \delta. \tag{36}$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from Theorem 3.3.

Proof. Fix the dual functions from Theorem 3.3. First use exponential coordinates. Unitarity and the one-parameter group derivative give

$$\langle H_A\Omega_n, (U_{\exp A} - I)\Omega_n\rangle = \int_0^1 \langle U_{\exp(-tA)}(H_A\Omega_n), G_A\Omega_n\rangle\, dt.$$

Replacing $U_{\exp(-tA)}(H_A\Omega_n)$ by $H_A\Omega_n$ in the integral gives the leading term $\frac{1}{2}\mathbb{E}H_AS_A = \frac{1}{2}\|A\|_F^2$. For a vector in the domain of G_A, the fundamental theorem of calculus gives $\|(U_{\exp(-tA)} - I)v\|_2 \le t\|G_Av\|_2$. Thus Theorem 3.4 and the Fisher upper bound imply

$$\left|\langle H_A\Omega_n, (U_{\exp A} - I)\Omega_n\rangle - \tfrac{1}{2}\|A\|_F^2\right| \le \frac{C_1\sqrt{C_f}}{4}\|A\|_F^3.$$

Cauchy–Schwarz and (30) now give, for all sufficiently small $\|A\|_F$,

$$h((\exp A)_{\#}\mu_n, \mu_n) \ge \frac{1}{4C_0}\|A\|_F.$$

For $\|B\|_{\mathrm{op}} < 1/2$, the real power-series logarithm $A = \log(I+B)$ exists and satisfies

$$\|A - B\|_F \le \sum_{k\ge 2} \frac{\|B\|_{\mathrm{op}}^{k-1}}{k}\|B\|_F \le C\|B\|_{\mathrm{op}}\|B\|_F.$$

Its norm is therefore uniformly comparable to $\|B\|_F$ in a fixed small chart. This proves the lower bound for $I + B$; the upper bound follows from Theorem 3.2. □

The preceding pairing is a half-density pairing: $\int H_A\sqrt{d\mu_n\, d((\exp A)_{\#}\mu_n)} - \int H_A\, d\mu_n$. It provides the uniform local certificate used below. The tail-symmetry argument in Section 5.2 is what lets this local certificate be applied to arbitrary output marginals without requiring a separate variance bound for H_A under every mixed output law.

3.4 Uniform coercivity modulo the exact linear stabilizer

The local estimate is the only part of this section needed for the infinite-dimensional classification. We state the global finite-dimensional form here because it is the natural finite-dimensional conclusion, but its proof is postponed until after the main theorem. The reason is that the global separation argument uses the infinite-dimensional classification as a compactness principle.

Theorem 3.6 (Dimension-free global Hellinger coercivity)**.** *Assume* (3)*. Let $\mathcal{P}_n$ denote the allowed signed-permutation matrices in dimension n, and set*

$$d_n(T) = \min_{Q \in \mathcal{P}_n} \|T - Q\|_F. \tag{37}$$

For every $0 < a \le b < \infty$ there are constants $c = c(f, a, b) > 0$ and $C = C(f, a, b) < \infty$ such that, for every n and every $T \in \mathrm{GL}(n, \mathbb{R})$ whose singular values lie in $[a, b]$,

$$c \min\{d_n(T)^2, 1\} \le h(\mu_n, T_\# \mu_n)^2 \le C \min\{d_n(T)^2, 1\}. \tag{38}$$

The local regime of (38) is exactly Theorem 3.5, after right multiplication by a member of $\mathcal{P}_n$. The uniform separation away from that local chart is a global statement. Its proof, given in Section 5.5, uses the infinite-dimensional equivalence theorem as a compactness device: a hypothetical sequence of distant near-invariances is assembled into one block-diagonal equivalence, which the Hilbert–Schmidt classification rules out. The global estimate is therefore a consequence of the main theorem and is not used in its proof.

4 Measurable Realization and Dichotomy

The sample space is $\mathbb{R}^{\mathbb{N}}$, not ℓ^2, so a bounded operator on ℓ^2 does not act pointwise by a naively convergent matrix product. This section constructs the required measurable versions. Then it identifies the global measure class through likelihood-ratio martingales and a dense translation group.

4.1 Measurable linear actions and countable-orbit realizations

Throughout this subsection, only $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ for $X \sim \rho$ are needed. Recall $\mu = \rho^{\otimes \mathbb{N}}$ and $\mu_n = \rho^{\otimes n}$. The sample sequence is not an ℓ^2-valued random variable. So we first specify how an operator on ℓ^2 acts on its law. The point is not formalism: the inverse action, tail exchanges, and likelihood cocycles used later must all hold on compatible measurable versions.

Lemma 4.1 (Measurable realization on each countable operator orbit)**.** *For every $T \in \mathcal{B}(H)$, there is a Borel map $\Phi_T : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \ge 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \le C_\lambda \|a\|_2^2 \quad \textit{for every finitely supported } a, \tag{39}$$

then $(\Phi_T)_\#\lambda$ also satisfies this condition. Its constant is at most $C_\lambda\|T\|_{\mathrm{op}}^2$. Also

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \qquad (S, T \in \mathcal{B}(H)). \tag{40}$$

For each fixed countable subgroup $\Gamma \leq \mathrm{GL}(H)$, there is a Γ-invariant Borel set $D \subset \mathbb{R}^{\mathbb{N}}$ of full measure for every λ satisfying (39). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ-orbit laws, including singular orbit laws.

Proof. For $a \in \ell^2$, the finite sums $\sum_{j\leq n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a, by (39). Choose

$$n_k(a) = \min\left\{ n \geq k : \sum_{j>n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define $\widehat{L}_a(x)$ to be the limit of $\sum_{j\leq n_k(a)} a_j x_j$ when it exists, and zero otherwise. The estimate

$$\left\| \sum_{j\leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C_\lambda 2^{-4k}$$

and Borel–Cantelli show that $\widehat{L}_a = L_a$, λ-almost surely. This one definition therefore works for every law satisfying (39); it does not depend on knowing whether that law is equivalent to μ. Set $\Phi_T(x)_i = \widehat{L}_{T^*e_i}(x)$. The construction is Borel, and the same is true jointly in the matrix coefficients of T and in x, since each tail norm and the integer choice n_k is Borel.

For finitely supported b, the finite sum of output coordinates equals L_{T^*b} almost surely. This proves the asserted second-moment bound for the pushforward. For the i-th row of S, finite output sums satisfy

$$\sum_{j\leq n} S_{ij}\widehat{L}_{T^*e_j} = L_{T^*P_nS^*e_i} \longrightarrow L_{T^*S^*e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law has the required second-moment bound. Uniqueness of the limit proves (40).

For a countable Γ, let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every law under consideration. Put

$$D = \bigcap_{T\in\Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies the same second-moment condition, so $\lambda(D) = 1$. The composition identities imply that D is invariant and that $\Phi_{T^{-1}}$ is the inverse of Φ_T there. This proves the final assertion. □

Write $\nu_T = (\Phi_T)_\#\mu$. For every finite or countable list of operators used in the proof, the preceding lemma supplies a common carrier on which the required action and inverse identities hold. This is all the later Hellinger identities need. No single pointwise action of all of $\mathrm{GL}(H)$ on one universal conull set is required. The subsequences in its proof are also essential. A covariance bound gives convergence of arbitrary row sums in L^2. That is not enough for almost-sure convergence of all ordinary partial sums under a correlated orbit law. On the original independent product law, the versions agree with the usual almost-sure row sums.

4.2 Rational translation invariance and the zero–one dichotomy

The main theorem will decide equivalence by proving either absolute continuity or its failure. The next lemma removes any intermediate case: once finite-dimensional densities are positive and ℓ^2-translations are quasi-invariant, every bounded invertible orbit is either equivalent to μ or singular to it. This lets later arguments prove one-sided statements and then upgrade them to the full alternative.

Lemma 4.2 (Dichotomy for every bounded invertible orbit)**.** *For every $T \in \mathrm{GL}(H)$,*

$$\nu_T \sim \mu \quad or \quad \nu_T \perp \mu. \tag{41}$$

This conclusion uses only positivity of f, its centered finite second moment, and $\sqrt{f} \in W^{1,2}(\mathbb{R})$. In particular, it holds under (3).

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - \mathsf{A}(f, f(\cdot - a)) = \tfrac{1}{2}\|g(\cdot - a) - g\|_2^2 \le \tfrac{1}{2}\|g'\|_2^2 a^2$$

and Kakutani's theorem imply $(x \mapsto x + a)_{\#}\mu \sim \mu$ for every $a \in \ell^2$. The row realizations satisfy

$$\Phi_T(x + T^{-1}a) = \Phi_T(x) + a \quad \mu\text{-almost surely;}$$

this follows directly by adding the absolutely convergent scalar row pairing with the ℓ^2 vector $T^{-1}a$. Consequently, ν_T is also quasi-invariant under every ℓ^2 translation.

Let $\mathcal{A}$ be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q},\ j \ge 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by $\mathcal{A}$. Indeed, an event invariant modulo μ under all rational finite-coordinate translations has, by positivity of the finite-dimensional densities, almost-everywhere constant sections in each finite coordinate block. It is therefore a tail event modulo μ, and has probability zero or one. Here one uses the elementary fact that a measurable function on $\mathbb{R}^n$ invariant almost everywhere under rational translations is constant almost everywhere, obtained, for example, by convolution with smooth approximate identities. For ν_T, pull an invariant event back by Φ_T; invariance under qTe_j becomes invariance under qe_j, giving the same zero–one conclusion.

Write $\nu_T = \nu_a + \nu_s$ for its Lebesgue decomposition relative to μ. Choose a Borel μ-null set N_0 supporting ν_s, and let $N = \bigcup_{a\in\mathcal{A}}(N_0 + a)$. Quasi-invariance of μ makes N a μ-null set; it is exactly $\mathcal{A}$-invariant and still supports ν_s. Thus $\nu_T(N) = \nu_s(\mathbb{R}^{\mathbb{N}})$, and ergodicity of ν_T makes this number zero or one. The latter case gives singularity. In the former case, write $\nu_T = Z\mu$. Quasi-invariance of both measures implies that $\{Z > 0\}$ is invariant modulo μ under $\mathcal{A}$. Its μ-measure is positive because $\int Z\, d\mu = 1$, and ergodicity of μ makes that measure one. This gives equivalence. □

4.3 Finite-marginal likelihood ratios and Hellinger energy

The classification is proved through finite-dimensional shadows. This lemma records how those shadows converge back to the infinite product measure: their likelihood ratios form martingales, and their Hellinger affinities decrease to the full affinity. Thus bounded projected energy is exactly the projective signal of equivalence, not merely a finite-dimensional necessary condition.

Lemma 4.3 (Projected Hellinger energy and finite-marginal likelihood ratios)**.** *Assume $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. For $T \in \mathrm{GL}(H)$, every finite output law $(\pi_n)_{\#}\nu_T$ is equivalent to μ_n. Writing $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, define*

$$\begin{gathered} \ell_n(x) = \frac{d[(\pi_n)_{\#}\nu_T]}{d\mu_n}(\pi_n x), \qquad E_n(T) = -2\log \mathbb{E}_\mu \sqrt{\ell_n}, \\ \mathsf{A}(\mu_n, (\pi_n)_{\#}\nu_T) = \mathbb{E}_\mu \sqrt{\ell_n}. \end{gathered} \tag{42}$$

Then (ℓ_n) is a nonnegative mean-one μ-martingale. We have

$$\mathsf{A}(\mu_n, (\pi_n)_{\#}\nu_T) \downarrow \mathsf{A}(\mu, \nu_T), \qquad \nu_T \sim \mu \Longleftrightarrow \sup_n E_n(T) < \infty. \tag{43}$$

Otherwise $E_n(T) \uparrow \infty$. Then $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density Z_T, then

$$\ell_n = \mathbb{E}_\mu[Z_T \mid \mathcal{F}_n] \longrightarrow Z_T \qquad \text{in } L^1(\mu), \tag{44}$$

$$\ell_n^{-1} = \mathbb{E}_{\nu_T}[Z_T^{-1} \mid \mathcal{F}_n] \longrightarrow Z_T^{-1} \qquad \text{in } L^1(\nu_T). \tag{45}$$

In particular, both likelihood-ratio martingales are uniformly integrable in their respective reference measures.

Proof. The first n rows of T are linearly independent, since T^* is injective. Some n input columns therefore form an invertible $n \times n$ minor A. Split the input into these coordinates and their complement. The first n outputs have the form $AU + W$, where $U \sim \mu_n$ and W is independent of U. Conditioning on W writes their density as a mixture of translates of the strictly positive density of AU. This mixture is positive and finite almost everywhere, proving the finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $\ell_\infty = d(\nu_T)_a/d\mu$, the density of the absolutely continuous part of ν_T. To see the identification explicitly, Fatou's lemma on cylinder sets gives $\ell_\infty \mu \le \nu_T$, first on the cylinder algebra and then on the generated sigma-field. Conversely, if $(\nu_T)_a = Z\mu$, then $\mathbb{E}_\mu[Z \mid \mathcal{F}_n] \le \ell_n$, so martingale convergence gives $Z \le \ell_\infty$. The two inequalities identify the limit. Since $\mathbb{E}_\mu(\sqrt{\ell_n})^2 = 1$, the square roots are uniformly integrable in $L^1(\mu)$. They thus converge in L^1 to $\sqrt{\ell_\infty}$. Conditional Jensen's inequality gives the monotonicity of their expectations, and consequently

$$\lim_n \mathsf{A}(\mu_n, (\pi_n)_{\#}\nu_T) = \int \sqrt{\ell_\infty}\, d\mu = \mathsf{A}(\mu, \nu_T).$$

The dichotomy in Theorem 4.2 now proves (43) and the singular alternative.

Under equivalence, the forward conditional-expectation identity is the definition of a finite-marginal density. The reverse identity follows because for $B \in \mathcal{F}_n$,

$$\int_B \ell_n^{-1}\, d\nu_T = \mu(B) = \int_B Z_T^{-1}\, d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. $\square$

By data processing, the supremum of the projected Hellinger energies is unchanged if one takes all finite coordinate marginals: the initial segments are cofinal among finite coordinate sets. Thus the criterion does not depend on the enumeration of the coordinates.

5 Proof of the Measure Equivalence Theorem

We now prove the measure-class classification. Sufficiency constructs a stable infinite-dimensional likelihood ratio from finite-coordinate Hilbert–Schmidt approximants. Necessity proves that any non-Hilbert–Schmidt tail deviation leaves a Hellinger trace that cannot be hidden by dense mixing. The global finite-dimensional coercivity theorem is proved at the end, after the classification supplies the required compactness principle.

5.1 Sufficiency of Hilbert–Schmidt perturbations

We first prove the constructive direction of the classification. A Hilbert–Schmidt perturbation can be approximated by finite-coordinate changes whose Hellinger increments are summable. The proposition turns this finite-dimensional Cauchy property into an actual infinite-dimensional likelihood ratio.

Proposition 5.1 (Hilbert–Schmidt sufficiency and likelihood ratio convergence)**.** *Assume* (15)*, let $\mu = \rho^{\otimes\mathbb{N}}$, where $\rho(dx) = f(x)\,dx$, and let $T = I + K \in \mathrm{GL}(H)$ with $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. For the eventually invertible operators $T_n = I + P_nKP_n$, their likelihood ratios $Z_{T_n} = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive likelihood ratio $Z_T = d\nu_T/d\mu$, and $\sqrt{Z_{T_n}} \to \sqrt{Z_T}$ in $L^2(\mu)$. The inverse approximants satisfy the same conclusion:*

$$\frac{d\nu_{T_n^{-1}}}{d\mu} \longrightarrow \frac{d\nu_{T^{-1}}}{d\mu} \quad \text{in } L^1(\mu).$$

For $S, T \in \mathrm{GL}(H)$ with $S - I, T - I \in \mathcal{S}_2$, the likelihood ratios obey, on the common measurable carrier of their countable operator group,

$$Z_{ST}(y) = Z_S(y)Z_T(\Phi_{S^{-1}}y), \tag{46}$$

$$Z_T(y)Z_{T^{-1}}(\Phi_{T^{-1}}y) = 1. \tag{47}$$

If Q is an exact linear stabilizer of μ, the same equivalence holds whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is invertible.

Proof. Since $P_nKP_n \to K$ in Hilbert–Schmidt norm, $T_n \to T$ in operator norm. The approximants are therefore eventually invertible. Their inverse norms are uniformly bounded. If A_n is the first n-coordinate block of T_n, then

$$Z_{T_n}(x) = |\det A_n|^{-1} \prod_{i=1}^{n} \frac{f((A_n^{-1}\pi_n x)_i)}{f(x_i)}. \tag{48}$$

These densities are positive almost everywhere. For all sufficiently large m, n, the operator $T_n^{-1}T_m - I$ acts in a finite coordinate block and has operator norm at most $1/2$. Hellinger invariance under the finite invertible map T_n^{-1} and (27) give

$$\|\sqrt{Z_{T_m}} - \sqrt{Z_{T_n}}\|_2 \le \sqrt{C_f}\,\|T_n^{-1}\|_{\mathrm{op}}\|T_m - T_n\|_{\mathcal{S}_2}. \tag{49}$$

Thus the square roots are Cauchy in $L^2(\mu)$. They have a nonnegative limit r of norm one. Cauchy–Schwarz also gives

$$\|Z_{T_m} - Z_{T_n}\|_1 \le 2\|\sqrt{Z_{T_m}} - \sqrt{Z_{T_n}}\|_2, \qquad Z_{T_n} \longrightarrow r^2 \quad \text{in } L^1(\mu). \tag{50}$$

For each fixed finite set of output coordinates, the row-series realizations $\Phi_{T_n}\mathbf{X}$ converge to $\Phi_T\mathbf{X}$ in L^2, where $\mathbf{X} = (X_j)_{j\geq 1}$ has law μ. Bounded continuous cylinder tests identify $r^2\mu$ with ν_T.

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\mathrm{op}}\|T_n - T\|_{\mathcal{S}_2}\|T^{-1}\|_{\mathrm{op}}.$$

Apply the same argument to these finite-coordinate approximants. This gives $\nu_{T^{-1}} \ll \mu$. The measurable row-series construction in Theorem 4.1 requires only the centered finite second moment. On its countable-group carrier it gives

$$\mu = (\Phi_T)_\#\nu_{T^{-1}} \ll (\Phi_T)_\#\mu = \nu_T.$$

This proves equivalence and strict positivity of r^2. Change of variables on the same carrier proves the cocycle and inverse identities. Finally, $\nu_Q = \mu$ and $T = (TQ^{-1})Q$ give the stabilizer assertion. □

5.1.1 The orthogonal case requires only location Fisher information

For orthogonal transformations the assumptions can be reduced to

$$f > 0 \text{ almost everywhere}, \qquad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \qquad \mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1. \tag{51}$$

In particular, $xg' \in L^2$ is not required. The cutoff identities $\mathbb{E}s(X) = 0$ and $\mathbb{E}[Xs(X)] = 1$ still follow from $J < \infty$ and the second moment. For a skew-symmetric matrix A, the score has only off-diagonal terms and

$$\mathbb{E}S_A^2 = (J-1)\|A\|_F^2. \tag{52}$$

The generator formula remains valid. Only $x_j\partial_i\Omega_n$ with $i \neq j$ occur. One can justify it by radial cutoff and radial mollification. Those operators commute with the orthogonal one-parameter group.

Proposition 5.2 (Orthogonal local bounds and Hilbert–Schmidt sufficiency)**.** *Under* (51)*, every orthogonal U on ℓ^2 with $U - I \in \mathcal{S}_2$ satisfies $\nu_U \sim \mu$. Its likelihood ratios have the same L^1 and square-root L^2 convergence along finite-coordinate orthogonal approximants, as do the inverse likelihood ratios. If f is non-Gaussian, there also exist dimension-free $c, C, \delta > 0$ such that*

$$c\|V - I\|_F \leq h(V_\#\mu_n, \mu_n) \leq C\|V - I\|_F \quad (V \in O(n),\ \|V - I\|_F \leq \delta). \tag{53}$$

Proof. For a skew-symmetric A, integration along $\exp(tA)$ gives

$$h((\exp A)_\#\mu_n, \mu_n) \leq \tfrac{1}{2}\sqrt{J-1}\,\|A\|_F. \tag{54}$$

If f is non-Gaussian, the construction of the two dual functions p, q in Theorem 3.3 uses only $s(X), X \in L^2$ and $J - 1 > 0$. On skew-symmetric matrices use $H_A = \sum_{i\neq j} A_{ij}p(X_i)q(X_j)$, with no diagonal function. The score S_B likewise has no diagonal term. Consequently the proof of Theorem 3.4 contains only bounded functions times $s(X)$ or X, and proves the same estimate under (51). The half-density pairing argument gives the local lower bound in exponential coordinates. For an orthogonal V sufficiently close to I, its power-series logarithm is real and skew-symmetric, with $\|\log V\|_F$ comparable to $\|V - I\|_F$. This proves (53).

For completeness, finite-coordinate orthogonal approximants can be constructed without applying a nonorthogonal path estimate. Set $T_n = I + P_n(U - I)P_n$ and, for large n, take its polar factor

$$V_n = T_n(T_n^* T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first n coordinates. Since $T_n \to U$ in Hilbert–Schmidt norm,

$$\|T_n^* T_n - I\|_{\mathcal{S}_2} \le (\|T_n\|_{\mathrm{op}} + 1)\|T_n - U\|_{\mathcal{S}_2} \longrightarrow 0.$$

The smallest singular values of T_n are bounded below by some $a > 0$. Functional calculus for $|T_n|$ then gives

$$\|V_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \le \frac{\|T_n^* T_n - I\|_{\mathcal{S}_2}}{1 + a}.$$

Thus $V_n \to U$ in Hilbert–Schmidt norm. For sufficiently large n, m, $V_n^* V_m$ is close to the identity and has a small skew-symmetric logarithm. Applying (54) to this logarithm proves that the square roots of the likelihood ratios are Cauchy. The L^1 limits of the likelihood ratios are identified by the same cylinder tests as in Theorem 5.1. Apply the argument also to $V_n^{-1} \to U^{-1}$; the measurable inverse action then proves equivalence and the asserted convergence. □

5.2 Necessity of Hilbert–Schmidt perturbations

The finite-dimensional lower bound must also apply to perturbations whose rows are not finitely supported. The next lemma transfers local coercivity from coordinate blocks to genuine Hilbert–Schmidt operators by approximation and likelihood-ratio convergence. It is the quantitative tool used later to turn many small tail mismatches into a contradiction.

Lemma 5.3 (The local bound for Hilbert–Schmidt operators). *Let f be non-Gaussian and satisfy (15). There are constants $c, C, \delta > 0$ such that every $K \in \mathcal{S}_2(\ell^2)$ with $\|K\|_{\mathcal{S}_2} \le \delta$ satisfies*

$$c\|K\|_{\mathcal{S}_2} \le h(\nu_{I+K}, \mu) \le C\|K\|_{\mathcal{S}_2}. \tag{55}$$

The constants can be chosen with $\delta < 1$, so $I + K$ is automatically invertible. Under the weaker assumptions (51), the same conclusion holds for orthogonal $I + K$ whenever f is non-Gaussian. This includes finite-rank perturbations whose ranges have infinite coordinate support.

Proof. In the general linear case, take $K_n = P_n K P_n$ and apply (36) to its finite coordinate block. Its untouched tail factors contribute affinity one, so the finite-dimensional Hellinger distance is also $h(\nu_{I+K_n}, \mu)$. The likelihood ratio convergence in Theorem 5.1 gives

$$h(\nu_{I+K_n}, \mu) = \left\| \sqrt{\frac{d\nu_{I+K_n}}{d\mu}} - 1 \right\|_2 \longrightarrow \left\| \sqrt{\frac{d\nu_{I+K}}{d\mu}} - 1 \right\|_2 = h(\nu_{I+K}, \mu).$$

Also $\|K_n\|_{\mathcal{S}_2} \to \|K\|_{\mathcal{S}_2}$. Passing to the limit proves both bounds.

In the orthogonal case, use the finite-coordinate polar approximants V_n and their likelihood ratio convergence from Theorem 5.2. They satisfy $V_n \to I + K$ in Hilbert–Schmidt norm. Choose the radius in this assertion to be half the finite-dimensional radius in (53). Then all sufficiently large V_n lie in that Frobenius ball, and

$$h(\nu_{V_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \qquad \|V_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Pass to the limit in (53). □

5.2.1 Characteristic-function rigidity and arbitrary mixing

Necessity begins by identifying what a single output row can look like if its law is close to the original coordinate law. This is a one-dimensional rigidity problem for linear combinations of independent copies of X. The characteristic-function argument below says that, for a non-Gaussian coordinate law, such a row must asymptotically be one signed coordinate.

Throughout this subsection, X_j are independent copies of a real random variable X with law ρ and

$$\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1,$$

and ρ is non-Gaussian. The characteristic-function arguments below require no moments of order greater than two. Recall

$$\phi(t) = \mathbb{E}e^{itX}, \qquad \mu = \rho^{\otimes\mathbb{N}}.$$

The group $\mathcal{P}$ consists of coordinate permutations whose row signs preserve ρ. This definition applies even when ρ has no density.

Lemma 5.4 (A product approximation for small coefficients)**.** *There is a nonnegative function η, with $\eta(\delta) \to 0$ as $\delta \downarrow 0$, such that, for every finite or countable real family (b_j) with $\sum_j b_j^2 < \infty$,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2 \right) \right| \leq \eta(\delta) t^2 \sum_j b_j^2 \quad \textit{if} \quad \sup_j |tb_j| \leq \delta. \tag{56}$$

For a countable family, the product denotes the characteristic function of the L^2 sum $\sum_j b_j X_j$.

Proof. The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\eta(\delta) := \sup_{0<|u|\leq\delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0, \qquad \eta(0) := 0.$$

Both $\phi(u)$ and $e^{-u^2/2}$ have modulus at most one. Telescoping the difference of their finite products therefore bounds it by the sum of the differences of the factors, proving (56) for a finite family. For a countable family, pass to the limit through finite partial sums in L^2 and hence in distribution. This argument does not take a logarithm of the characteristic function. ☐

Lemma 5.5 (Qualitative rigidity of a unit row)**.** *Let $a^{(n)} \in \ell^2$ satisfy $\|a^{(n)}\|_2 = 1$. If*

$$\mathrm{Law}\left(\sum_j a_j^{(n)} X_j \right) \Longrightarrow \rho,$$

then

$$\inf_{\substack{j\geq 1,\ \varepsilon\in\{-1,1\} \\ \mathrm{Law}(\varepsilon X)=\rho}} \|a^{(n)} - \varepsilon e_j\|_2 \longrightarrow 0. \tag{57}$$

Proof. We first describe the limits of such coefficient arrays. Permuting coefficients does not change the law of their sum. Arrange each row in nonincreasing order of absolute value, retaining its signs and padding with zeros if necessary. Its jth coefficient then has modulus at most $j^{-1/2}$. From any subsequence we may extract a further subsequence on which each coefficient converges, say $a_j^{(n)} \to a_j$. Fatou's lemma gives

$$q := \sum_j a_j^2 \leq 1.$$

For a fixed m, apply Theorem 5.4 to the coefficients with $j > m$. For every fixed t, the error in replacing their characteristic function by

$$\exp\left[-\frac{t^2}{2}\left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

is at most $t^2\eta(|t|/\sqrt{m+1})$, uniformly in n. Fix any t. First let $n \to \infty$, and then let $m \to \infty$. The limiting characteristic function is

$$e^{-(1-q)t^2/2} \prod_j \phi(a_j t). \tag{58}$$

Equivalently, the limiting law is that of

$$\sum_j a_j X_j + \sqrt{1-q}\, Z,$$

where Z is standard Gaussian and independent of the X_j.

Under the convergence hypothesis this limiting law is the law of X. Suppose that no $|a_j|$ is one. If all a_j vanish, (58) already makes X Gaussian. Otherwise the maximum $\theta := \max_j |a_j|$ exists, since $(a_j) \in \ell^2$, and satisfies $0 < \theta < 1$. Iterating the characteristic-function identity (58) k times gives

$$\phi(t) = e^{-(1-q^k)t^2/2} \prod_{\alpha \in \mathbb{N}^k} \phi(tb_\alpha), \qquad b_\alpha = a_{\alpha_1} \cdots a_{\alpha_k}. \tag{59}$$

Here

$$\sum_\alpha b_\alpha^2 = q^k, \qquad \sup_\alpha |b_\alpha| \leq \theta^k.$$

The identity may be obtained by substituting independent copies into the L^2 series; at each finite depth all series have summable coefficient squares. Applying Theorem 5.4 to the last product in (59) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 q^k \eta(|t|\theta^k) \longrightarrow 0.$$

This again makes X Gaussian. This is a contradiction. Thus some $a_j = \varepsilon \in \{-1, 1\}$; all the remaining coefficients and the independent Gaussian summand vanish. The limiting identity then says $\mathrm{Law}(\varepsilon X) = \rho$, so the sign preserves ρ.

To conclude, suppose (57) failed. Take a subsequence whose distance from all the indicated unit vectors is bounded below. The preceding extraction produces a coefficient converging to an allowed sign of modulus one. Since the row norms are one, its squared distance from that signed unit vector tends to zero:

$$\|a^{(n)} - \varepsilon e_j\|_2^2 = 2 - 2\varepsilon a_j^{(n)} \longrightarrow 0.$$

Undoing the coefficient permutations gives the required contradiction. $\square$

Lemma 5.6 (Uniform invariance under tail symmetries)**.** *Let μ be the i.i.d. product law and let $\nu \ll \mu$. Let $\mathcal{T}_N$ consist of the members of $\mathcal{P}$ that change only finitely many coordinates and fix coordinates $1, \dots, N$. Then*

$$\sup_{S \in \mathcal{T}_N} h(\nu, S_{\#}\nu) \longrightarrow 0. \tag{60}$$

Also, the one-coordinate marginals of ν converge to the common coordinate law of μ, uniformly over coordinates greater than N.

Proof. Put $\ell = d\nu/d\mu$ and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. The martingale convergence theorem gives $\|\ell - \ell_N\|_{L^1(\mu)} \to 0$. Every $S \in \mathcal{T}_N$ preserves μ and satisfies $\ell_N \circ S^{-1} = \ell_N$. Thus

$$h(\nu, S_{\#}\nu)^2 \le \|\ell - \ell \circ S^{-1}\|_{L^1(\mu)} \le 2\|\ell - \ell_N\|_{L^1(\mu)},$$

which proves (60). For $i > N$, the marginal density is $\mathbb{E}_\mu[\ell \mid x_i]$. Independence and $\mathbb{E}_\mu \ell_N = 1$ imply $\mathbb{E}_\mu[\ell_N \mid x_i] = 1$, so

$$\|\mathbb{E}_\mu[\ell \mid x_i] - 1\|_{L^1(\mu)} \le \|\ell - \ell_N\|_{L^1(\mu)}.$$

This proves the uniform marginal assertion. $\square$

Lemma 5.7 (Compactness of the covariance defect)**.** *Let T be bounded on ℓ^2. If $\nu_T \ll \mu$, then*

$$TT^* - I \quad \textit{is compact.}$$

If T is boundedly invertible, $T^*T - I$ is compact as well.

Proof. We first record uniform integrability of the squares of the linear sums

$$Y_b := \sum_j b_j X_j, \qquad \|b\|_2 \le B, \tag{61}$$

for each fixed $B < \infty$. Let

$$X^{(M)} = X\mathbf{1}_{\{|X| \le M\}} - \mathbb{E}[X\mathbf{1}_{\{|X| \le M\}}], \qquad Y_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables $X^{(M)}$ are bounded and centered, and converge to X in L^2. Independence gives

$$\sup_{\|b\|_2 \le B} \|Y_b - Y_b^{(M)}\|_2 \le B\|X - X^{(M)}\|_2 \longrightarrow 0.$$

For fixed M, expanding the fourth moment of a sum of independent centered bounded variables gives

$$\sup_{\|b\|_2 \le B} \mathbb{E}|Y_b^{(M)}|^4 < \infty.$$

The bound first holds for finite sums; it also makes their truncations Cauchy in L^4, proving the countable version. Moreover,

$$\sup_{\|b\|_2 \le B} \left\|Y_b^2 - (Y_b^{(M)})^2\right\|_1 \le B^2\|X - X^{(M)}\|_2(\|X\|_2 + \|X^{(M)}\|_2) \longrightarrow 0.$$

Thus the squares in (61) are uniformly approximated in L^1 by uniformly integrable families, and are themselves uniformly integrable. This uses fourth moments of bounded truncations, not a fourth-moment assumption on X.

Write $\nu = \nu_T$, $\ell = d\nu/d\mu$, and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. For a finitely supported unit vector a supported on coordinates greater than N, put $F_a(x) = (\sum_i a_i x_i)^2$. Under μ, these are squares from (61) with $B = 1$; under ν, they are squares of $Y_{T^* a}$ under μ, with $\|T^* a\|_2 \le \|T\|_{\mathrm{op}}$. They are therefore uniformly integrable under both laws. The truncated statistic $F_a \wedge L$ depends only on tail coordinates. Independence yields

$$|\mathbb{E}_\nu(F_a \wedge L) - \mathbb{E}_\mu(F_a \wedge L)| \le L\|\ell - \ell_N\|_1.$$

First let $N \to \infty$ with fixed L. Then let $L \to \infty$. Uniform integrability gives

$$\sup_{\substack{\|a\|_2=1,\ a \text{ finitely supported} \\ \operatorname{supp} a \subset \{N+1, N+2, \dots\}}} |\langle (TT^* - I)a, a\rangle| \longrightarrow 0.$$

Since $TT^* - I$ is self-adjoint and bounded, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\mathrm{op}} \longrightarrow 0.$$

The complementary rows and columns form a finite-rank operator. Hence $TT^* - I$ is a norm limit of finite-rank operators and is compact. For invertible T, use $T^*T - I = T^{-1}(TT^* - I)T$. □

We now assemble the necessity argument. The preceding row rigidity gives only qualitative matching of far-out rows with signed coordinates. To obtain a Hilbert–Schmidt conclusion, those row errors must be summable. The proof forces summability by swapping carefully chosen tail coordinates: if the errors were not summable, the swaps would create a small finite-rank perturbation that local coercivity can detect, while tail invariance makes it asymptotically invisible.

Lemma 5.8 (Hilbert–Schmidt necessity under a second-moment hypothesis)**.** *Assume that the common law has a positive density f, that $g = \sqrt{f} \in W^{1,2}(\mathbb{R})$, and that $xg' \in L^2(\mathbb{R})$. If $T \in \mathrm{GL}(H)$ and $\nu_T \sim \mu$, then*

$$T - Q \in \mathcal{S}_2 \qquad \textit{for some } Q \in \mathcal{P}. \tag{62}$$

When T is orthogonal, the condition $xg' \in L^2$ can be omitted.

Proof. By Theorem 5.7, $\|T^* e_i\|_2 \to 1$. By Theorem 5.6, the law of $\sum_j (T^* e_i)_j X_j$ converges to ρ. Normalizing $T^* e_i$ to a unit vector changes this sum by a quantity tending to zero in L^2. Therefore Theorem 5.5 gives indices $\pi(i)$ and signs $\varepsilon_i \in \{-1, 1\}$ preserving ρ, outside a finite initial set, with

$$u_i := \varepsilon_i e_{\pi(i)}, \qquad \|T^* e_i - u_i\|_2^2 \longrightarrow 0. \tag{63}$$

The indices $\pi(i)$ are distinct sufficiently far out. Indeed, if $\pi(i) = \pi(j)$ for two such rows, then $\varepsilon_i T^* e_i - \varepsilon_j T^* e_j$ has arbitrarily small norm by (63), whereas bounded invertibility gives the lower bound $\sqrt{2}/\|T^{-1}\|_{\mathrm{op}}$. Hence we can fix N_0 so that π is injective on $i > N_0$.

There is also column approximation along this matching. Compactness of $T^*T - I$ and the fact that $\pi(i)$ leaves every finite set imply

$$\|T u_i\|_2^2 \longrightarrow 1.$$

Since $\langle Tu_i, e_i\rangle = \langle u_i, T^* e_i\rangle \to 1$, we obtain $\|Tu_i - e_i\|_2^2 \to 0$. Consequently

$$\eta_i := \|T^* e_i - u_i\|_2^2 + \|Tu_i - e_i\|_2^2 \longrightarrow 0. \tag{64}$$

We prove that $\sum_i \eta_i < \infty$ on this tail. By Theorem 5.3, there are $c_0, \delta_0 > 0$ such that

$$h(\mu, \nu_W) \geq c_0 \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \mathrm{GL}(H), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_0. \tag{65}$$

Suppose instead that $\sum_i \eta_i = \infty$. Choose $\delta > 0$ small enough that $3\|T^{-1}\|_{\mathrm{op}}\delta < \delta_0$. In any sufficiently remote tail, where $\eta_i \leq \delta^2$, choose a finite set E satisfying

$$\delta^2 \leq \sum_{i \in E} \eta_i \leq 2\delta^2. \tag{66}$$

Pair each $i \in E$ with a distinct auxiliary index k_i beyond E, and set $G = \{k_i : i \in E\}$. Since $\eta_i \to 0$, these auxiliary indices may be chosen with an arbitrarily small value of $\sum_{k \in G} \eta_k$, while E remains fixed.

Let S exchange each output pair (e_i, e_{k_i}), and let P exchange the aligned signed input pairs (u_i, u_{k_i}), acting identically on their orthogonal complements. Both are involutions in $\mathcal{P}$, so they preserve μ. Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \qquad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|_{\mathrm{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\|_{\mathrm{op}} \|ST - TP\|_{\mathcal{S}_2}. \tag{67}$$

These are finite-rank differences. Indeed $S - I$ and $P - I$ have finite rank. Hellinger invariance and $P_{\#}\mu = \mu$ give

$$h(\mu, \nu_W) = h(\nu_T, S_{\#}\nu_T). \tag{68}$$

All maps in use lie in a countable operator group, so Theorem 4.1 justifies this identity on a common carrier.

We now estimate $ST - TP$ in (67). Obtain T_0 from T by replacing each auxiliary row and column with its matched coordinate vector:

$$T_0^* e_k = u_k, \qquad T_0 u_k = e_k \qquad (k \in G).$$

On the orthogonal complements of these input and output coordinates, retain the corresponding block of T. These conditions uniquely determine T_0. Orthogonal block decomposition then gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} \left(\|T^* e_k - u_k\|_2^2 + \|Tu_k - e_k\|_2^2 \right) = \sum_{k \in G} \eta_k. \tag{69}$$

For completeness, the left side counts the squared entries of the auxiliary row and column errors once, while their intersection is counted twice on the right.

Order the output subspaces as E, its complement outside $E \cup G$, and G. Order the input subspaces as the aligned signed coordinates for E, their complement outside the aligned $E \cup G$, and the aligned coordinates for G. In these orthonormal decompositions,

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I \end{pmatrix}, \qquad ST_0 - T_0 P = \begin{pmatrix} 0 & -B & I - A \\ C & 0 & -C \\ A - I & B & 0 \end{pmatrix}.$$

The blocks B, C have finite-dimensional range or domain and are Hilbert–Schmidt. The sum of squared row and column errors on E is

$$\sum_{i \in E} (\|T_0^* e_i - u_i\|_2^2 + \|T_0 u_i - e_i\|_2^2) = 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2.$$

The displayed block matrix gives the exact identity

$$\|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 = 2(\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

and therefore

$$\begin{aligned} 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2 &\le \|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 \\ &\le 2(2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2). \end{aligned} \tag{70}$$

As the auxiliary error in (69) tends to zero, the displayed sum of squared errors tends to $\sum_{i \in E} \eta_i$. Also,

$$\|(ST - TP) - (ST_0 - T_0 P)\|_{\mathcal{S}_2} \le 2\|T - T_0\|_{\mathcal{S}_2} \longrightarrow 0.$$

By (66) and (70), we can choose actual finite auxiliary indices so that

$$\frac{\delta}{2} \le \|ST - TP\|_{\mathcal{S}_2} \le 3\delta.$$

It follows from (67) that

$$\frac{\delta}{2\|T\|_{\mathrm{op}}} \le \|W - I\|_{\mathcal{S}_2} \le 3\|T^{-1}\|_{\mathrm{op}} \delta < \delta_0.$$

The local bound (65) now forces $h(\mu, \nu_W) \ge c_0 \delta / (2\|T\|_{\mathrm{op}})$. On the other hand, S is supported in the original remote tail. Theorem 5.6 makes the right side of (68) uniformly smaller than this positive number when that tail is far enough out. This is the contradiction. Therefore $\sum_i \eta_i < \infty$.

Define Q_0 by the matched rows u_i after the fixed initial cutoff N_0, and by zero rows for $i \le N_0$. The summability just proved implies

$$\begin{aligned} \|T - Q_0\|_{\mathcal{S}_2}^2 &= \sum_{i \le N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \|T^* e_i - u_i\|_2^2 \\ &\le \sum_{i \le N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \eta_i < \infty. \end{aligned}$$

Therefore Q_0 is a compact perturbation of an invertible operator and is Fredholm of index zero. Its cokernel consists precisely of the first N_0 output coordinates. Its kernel is spanned by the unused input coordinates. There are exactly N_0 unused columns. Match them to the first N_0 rows with sign $+1$. This completes Q_0 to a single $Q \in \mathcal{P}$ and proves (62).

If T is orthogonal, both covariance defects vanish, and every operator $W = PT^*ST$ constructed above is orthogonal. Consequently only the orthogonal part of Theorem 5.3 is needed. That part uses location Fisher information alone, so the condition $xg' \in L^2$ may be omitted in this case. □

5.3 Exact identification of the linear stabilizer

The equivalence theorem needs the Sobolev assumptions. It constructs and controls likelihood ratios. Exact invariance is more rigid. It requires only the second moment.

Proposition 5.9 (Exact linear stabilizer under a second-moment hypothesis)**.** *For a centered, variance-one, non-Gaussian i.i.d. product law, the bounded invertible linear stabilizer is exactly $\mathcal{P}$. This assertion requires neither a density nor Fisher information.*

Proof. Every member of $\mathcal{P}$ preserves the product law. Conversely, if $\nu_T = \mu$, covariance gives $TT^* = I$. Since T is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as X. Applying Theorem 5.5 to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus $T \in \mathcal{P}$. □

5.4 Two-sided uniform integrability and completion of the proof

There are two likelihood ratio limits in the argument. The finite-coordinate approximants in Theorem 5.1 converge in $L^1(\mu)$. Their square roots converge in $L^2(\mu)$. Independently, once equivalence is known, the finite-marginal likelihood ratios and their reciprocals are the two conditional-expectation martingales in (44)–(45). This gives uniform integrability in both directions. It also rules out loss of mass at the projective limit.

Proof of Theorem 1.1 and Theorem 1.2. Necessity follows from Theorem 5.8. If $T - Q \in \mathcal{S}_2$ for $Q \in \mathcal{P}$, then $TQ^{-1} = I + (T - Q)Q^{-1}$ is invertible and differs from the identity by a Hilbert–Schmidt operator. By Theorem 5.1, $\nu_{TQ^{-1}} \sim \mu$. Since $Q_{\#}\mu = \mu$, this is $\nu_T \sim \mu$. The same proposition gives the finite-coordinate likelihood ratio and square-root limits stated in Theorem 1.2. The exact stabilizer is identified in Theorem 5.9. Finally, Theorems 4.2 and 4.3 give the finite-marginal affinity characterization and the singular alternative in Theorem 1.1, and both uniform-integrability statements in Theorem 1.2. □

Theorem 5.10 (Complete orthogonal criterion with only location Fisher information)**.** *Let $f > 0$ almost everywhere be a centered, variance-one, non-Gaussian density, and assume only $\sqrt{f} \in W^{1,2}(\mathbb{R})$ in addition to its second moment. For every orthogonal U on ℓ^2,*

$$\nu_U \sim \mu \quad \Longleftrightarrow \quad U - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}.$$

If this condition fails, $\nu_U \perp \mu$.

Proof. Necessity is Theorem 5.8. For sufficiency, write $UQ^{-1} = I + (U - Q)Q^{-1}$. This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply Theorem 5.2, and then use $Q_{\#}\mu = \mu$. The alternative follows from Theorem 4.2. □

5.5 Globalization of finite-dimensional coercivity

We now prove the global finite-dimensional statement announced in Section 3.4. This final argument is deliberately placed after the main theorem. It uses the infinite-dimensional classification to rule out a sequence of finite-dimensional almost-symmetries that stay a fixed distance from the finite stabilizer. In this way the Hilbert–Schmidt theorem supplies the compactness missing from a purely finite-dimensional proof.

Proof of Theorem 3.6. Fix $0 < a \le b < \infty$. We first prove uniform separation away from the stabilizer: for every $\varepsilon > 0$,

$$\inf \left\{ h(\mu_n, A_\# \mu_n)^2 : \begin{array}{l} n \ge 1,\ A \in \mathrm{GL}(n, \mathbb{R}), \\ s_{\min}(A) \ge a,\ s_{\max}(A) \le b, \\ d_n(A) \ge \varepsilon \end{array} \right\} > 0. \tag{71}$$

Suppose otherwise. Then there are dimensions n_k and matrices A_k in the displayed class with $h(\mu_{n_k}, (A_k)_\# \mu_{n_k}) \to 0$. Since

$$-2 \log \mathsf{A}(\mu_{n_k}, (A_k)_\# \mu_{n_k}) = -2 \log \left(1 - \tfrac{1}{2} h(\mu_{n_k}, (A_k)_\# \mu_{n_k})^2\right),$$

we may pass to a subsequence. Without changing notation, assume this sum of block energies is finite.

Decompose $H = \bigoplus_{k \ge 1} \mathbb{R}^{n_k}$ into consecutive coordinate blocks. Now define $A = \bigoplus_{k \ge 1} A_k$. The common singular-value bounds imply $A \in \mathrm{GL}(H)$. Each block law $(A_k)_\# \mu_{n_k}$ is equivalent to μ_{n_k} because $f > 0$ almost everywhere and A_k is invertible. The summability of the block energies gives

$$\prod_{k \ge 1} \mathsf{A}(\mu_{n_k}, (A_k)_\# \mu_{n_k}) > 0.$$

Kakutani's product theorem [10] therefore gives $\nu_A \sim \mu$. The already proved Theorem 1.1 therefore supplies $Q \in \mathcal{P}$ such that

$$D := A - Q \in \mathcal{S}_2(H). \tag{72}$$

Let Π_k be the orthogonal projection onto the k-th output block. The orthogonality of these blocks gives

$$\sum_k \|\Pi_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \qquad \text{hence} \qquad \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0.$$

If, for some output basis vector e_i in the k-th block, the signed basis vector $Q^* e_i$ lies outside the corresponding input block, then $A^* e_i$ and $Q^* e_i$ have disjoint supports. The lower singular value bound then yields

$$\|A^* e_i - Q^* e_i\|_2^2 = \|A_k^* e_i\|_2^2 + 1 \ge a^2 + 1. \tag{73}$$

For all sufficiently large k, this contradicts $\|\Pi_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$. Hence Q^* maps every output coordinate of each sufficiently large block into the same input block. Its restriction is injective between two sets of cardinality n_k. Hence it is a bijection. Denote the resulting member of $\mathcal{P}_{n_k}$ by Q_k. It follows that

$$d_{n_k}(A_k) \le \|A_k - Q_k\|_F = \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0,$$

contradicting $d_{n_k}(A_k) \ge \varepsilon$. This proves (71).

Let c_0, C_0, δ_0 be the constants in Theorem 3.5, decreasing δ_0 so that $0 < \delta_0 < 1$. Given T, choose a minimizer $Q \in \mathcal{P}_n$ in (37). Since $Q_\# \mu_n = \mu_n$,

$$h(\mu_n, T_\# \mu_n) = h(\mu_n, (TQ^{-1})_\# \mu_n), \qquad \|TQ^{-1} - I\|_F = d_n(T).$$

When $d_n(T) \le \delta_0$, the local theorem supplies the two-sided linear bounds for h. When $d_n(T) > \delta_0$, (71) with $\varepsilon = \delta_0$ supplies the lower bound, while $h^2 \le 2$ supplies the upper bound. Choose $c > 0$ no larger than c_0^2 and the positive infimum in (71) with $\varepsilon = \delta_0$, and take

$$C = \max\{C_0^2, 2\delta_0^{-2}\}.$$

These constants give (38) in all regimes. $\square$

6 Density Zeros and the Linear Equivalence Group

The positivity assumption in the main theorem has one precise job: it makes finite-coordinate likelihood ratios blind to support. Once the density has zeros, the Hilbert–Schmidt condition remains the right metric accumulation condition, but it is no longer the whole story. The missing datum is row-wise support compatibility. Each row of the operator, and for equivalence each row of the inverse, must land in the positivity set of the one-dimensional density almost surely.

This section gives that support-aware replacement first. Arbitrary zero sets lead to an exact zero-set row criterion; bounded coordinate laws force signed permutations with only the scalar reflection freedom allowed by the one-dimensional measure class; centered half-line laws force ordinary permutations. We then return to examples showing why the old positive-density dichotomy fails and how boundary behavior changes the summability threshold. The hypotheses are stated separately for each result; (3) is not a standing assumption.

6.1 The metric input when the density has zeros

The support-aware theorem still uses the same Fourier/Fisher mechanism for the metric part of the classification. The logarithm-free Fourier characterization in Theorem 2.1, and its one-rotation Hellinger lower bound, already require no density. For the stronger matrix estimates, the appropriate regularity condition is

$$\begin{gathered} f \ge 0, \quad \int f = 1, \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \text{ non-Gaussian}. \end{gathered} \tag{74}$$

Here g is defined on the whole real line, including by zero outside the support. Regularity only in the interior of the support is insufficient. For example, the uniform density has zero interior derivative but its square root has jumps at the boundary.

Proposition 6.1 (Local Fourier coercivity with zeros of the density)**.** *Under (74), the matrix Fisher decomposition, finite-dimensional Hellinger path bound, and dimension-free local Frobenius coercivity in Section 3 remain valid. Write $\mu = \rho^{\otimes \mathbb{N}}, \quad \rho(dx) = f(x)\,dx$. There are constants $c, C, \delta > 0$, with $\delta < 1$, such that*

$$c\|K\|_{\mathcal{S}_2} \le h(\mu, \nu_{I+K}) \le C\|K\|_{\mathcal{S}_2} \qquad (\|K\|_{\mathcal{S}_2} \le \delta). \tag{75}$$

Moreover, for every $T \in \mathrm{GL}(H)$, the following necessary condition survives:

$$\nu_T \ll \mu \quad \Longrightarrow \quad T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \tag{76}$$

For orthogonal transformations one may omit $xg' \in L^2$, retaining non-Gaussianity, $g \in W^{1,2}(\mathbb{R})$, and the centered variance-one normalization. These statements do not assert absolute continuity of ν_{I+K}.

Proof. A Sobolev function has weak derivative zero almost everywhere on its zero set. Define $s = -2g'/g$ on $\{g > 0\}$ and $s = 0$ elsewhere. Then $sg = -2g'$ almost everywhere on $\mathbb{R}$. The cutoff integration-by-parts identities, including $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and the formulas for J, κ, are therefore unchanged. Equality $J = 1$ still gives the global equation $g' = -xg/2$; it cannot produce a Gaussian truncated at a boundary. Similarly, $\kappa = 0$ would give $2xg' + g = 0$

globally and no nonzero square-integrable probability half-density. The finite real Fourier sums with dual moments and half-density generator estimates use these identities and L^2 regularity, so their proofs apply.

The infinite-dimensional limiting step needs a different interpretation. For $K \in \mathcal{S}_2$ with $I + K$ invertible, put $T_n = I + P_n K P_n$. The finite-dimensional path bound and invariance of Hellinger distance give, for sufficiently large m, n,

$$h(\nu_{T_n}, \nu_{T_m}) \le C \|T_n^{-1}\|_{\mathrm{op}} \|T_n - T_m\|_{\mathcal{S}_2}.$$

Choose a single probability measure dominating the countable family $\mu, \nu_{T_1}, \nu_{T_2}, \ldots$. Their square-root densities are Cauchy in its L^2 space. The nonnegative limit has norm one and defines a probability measure; convergence of finite output row sums identifies this measure with ν_{I+K}. This proves Hellinger convergence, not a density relative to μ. Passing the finite local bounds to this limit proves (75). For orthogonal operators, use the finite-coordinate polar approximants and skew-generator bound from Theorem 5.2 in the same way.

Finally, the proof of Theorem 5.8 uses absolute continuity only through Theorems 5.6 and 5.7, which assume $\nu_T \ll \mu$. Its other inputs are the moment-only characteristic-function rigidity, the measurable inverse action, and the local estimate for the finite-rank differences $PT^{-1}ST - I$. The estimate just established supplies this last input without any quasi-invariance assumption. The same tail matching and permutation argument therefore proves (76). □

For standardized Gamma shape α, the zero-extended square root is in $W^{1,2}(\mathbb{R})$ exactly when $\alpha > 2$. For an affine rescaling of a Beta(α, β) density the corresponding condition is $\alpha, \beta > 2$. These assertions follow by squaring the derivative of the boundary power $t^{(\alpha-1)/2}$. At shape one the square root instead has a jump, so its vanishing interior power derivative is not an exception. The support classifications below require none of this regularity and include every positive Gamma or Beta shape.

6.2 The linear equivalence group for arbitrary zero sets of the density

The first replacement theorem is the main support-aware form of the classification. The Hilbert–Schmidt alternative remains the metric part of the answer. The new condition is purely about null sets: no output coordinate may place positive probability in the zero set of the original density, and mutual equivalence requires the same condition after applying the inverse operator. The condition concerns the actual zero set of the density, rather than its closed support.

Theorem 6.2 (The linear equivalence group with density zeros)**.** *Assume (74), let $\rho(dx) = f(x)\,dx$, and put $\mathcal{U} = \{x \in \mathbb{R} : f(x) > 0\}$. For every $T \in \mathrm{GL}(H)$,*

$$\nu_T \ll \mu \quad \Longleftrightarrow \quad \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu\left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i. \end{cases} \tag{77}$$

Consequently the linear equivalence group of μ is

$$\begin{aligned}\mathcal{G}_\rho &:= \{T \in \mathrm{GL}(H) : \nu_T \sim \mu\} \\ &= \left\{ T \in \mathrm{GL}(H) : \begin{array}{c} T - Q \in \mathcal{S}_2(H) \textit{ for some } Q \in \mathcal{P}, \\ \mu\left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0, \\ \mu\left\{ x : \sum_j (T^{-1})_{ij} x_j \notin \mathcal{U} \right\} = 0 \quad \textit{for every } i \end{array} \right\}. \end{aligned} \tag{78}$$

This set is a subgroup of $\mathrm{GL}(H)$*. For orthogonal members of* $\mathcal{G}_\rho$*, the same criterion holds with* $T^{-1} = T^*$ *and without the scale condition* $xg' \in L^2$*.*

More precisely, whenever $T - Q \in \mathcal{S}_2(H)$ *for some* $Q \in \mathcal{P}$*, the Lebesgue decomposition of* ν_T *relative to* μ *is its restriction to* $\mathcal{U}^{\mathbb{N}}$ *plus its restriction to the complement: the former is absolutely continuous and the latter is singular.*

Proof. First suppose $T - Q \in \mathcal{S}_2$ and put $K = TQ^{-1} - I$. The exact invariance $\nu_Q = \mu$ gives $\nu_T = \nu_{I+K}$. For the eventually invertible finite-coordinate approximants $T_n = I + P_n K P_n$, the proof of Theorem 6.1 gives

$$h(\nu_{T_n}, \nu_T) \longrightarrow 0.$$

In particular their probabilities converge uniformly over measurable sets, by the Hellinger bound on total variation.

For each such n, the law ν_{T_n} has a Lebesgue density in the first n coordinates and retains the independent product law on the remaining coordinates. On $\mathcal{U}^n$, the density of μ_n is strictly positive. Fubini's theorem therefore gives

$$\nu_{T_n}|_{\mathcal{U}^{\mathbb{N}}} \ll \mu.$$

If B is any μ-null Borel set, then $\nu_{T_n}(B \cap \mathcal{U}^{\mathbb{N}}) = 0$ for all sufficiently large n. Total variation convergence implies $\nu_T(B \cap \mathcal{U}^{\mathbb{N}}) = 0$. Thus $\nu_T|_{\mathcal{U}^{\mathbb{N}}} \ll \mu$. Since $\mu(\mathcal{U}^{\mathbb{N}}) = 1$, the restriction to its complement is singular. This proves the asserted decomposition and

$$\nu_T \ll \mu \quad \iff \quad \nu_T(\mathcal{U}^{\mathbb{N}}) = 1 \quad \iff \quad \mu\left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i.$$

The necessary Hilbert–Schmidt condition in (76) now proves (77) in both directions.

If $\nu_T \sim \mu$, the measurable inverse action gives $\nu_{T^{-1}} \sim \mu$, so both row conditions in (78) are necessary. Conversely, $T - Q \in \mathcal{S}_2$ implies

$$T^{-1} - Q^{-1} = -T^{-1}(T - Q)Q^{-1} \in \mathcal{S}_2.$$

Apply (77) to T and T^{-1} to obtain $\nu_T \ll \mu$ and $\nu_{T^{-1}} \ll \mu$. The measurable inverse action then gives

$$\mu = (\Phi_T)_\# \nu_{T^{-1}} \ll (\Phi_T)_\# \mu = \nu_T,$$

which proves equivalence. For $S, T \in \mathcal{G}_\rho$, measurable composition yields $\nu_{ST} = (\Phi_S)_\# \nu_T \sim \nu_S \sim \mu$; the inverse argument already proved gives $T^{-1} \in \mathcal{G}_\rho$. Thus the displayed operator class is a group. For orthogonal transformations, repeat the restriction argument with the finite-coordinate orthogonal approximants from Theorem 5.2; their Hellinger convergence needs only location Fisher information. □

Thus density zeros do not introduce an additional global likelihood-ratio limit beyond the operator criterion. They introduce a row-by-row boundary test. Every nonzero row sum has an absolutely continuous scalar law: condition on all but one input with a nonzero coefficient. Thus the row conditions are unchanged if f is modified on a Lebesgue-null set. If $f > 0$ almost everywhere, all these conditions are automatic and (78) reduces to Theorem 1.1. For a density with gaps, the rows of T and T^{-1} must avoid those gaps almost surely. The contraction in Section 6.5 below satisfies the forward row condition but fails the inverse row condition, which is exactly why it gives strict absolute continuity rather than equivalence.

6.3 Boundary behavior of the density and summability criteria

The row test and the Hellinger energy measure different phenomena. Boundary compatibility decides whether the transformed law is absolutely continuous in either direction; Hellinger loss decides whether overlap accumulates to zero. The two block-rotation examples below separate these effects.

Boundary type	**One-block loss**	**What it controls**
Hard boundary, uniform square	linear in $\lvert\theta\rvert$	accumulated Hellinger loss; absolute continuity still fails for every nonzero boundary-crossing block
Smooth vanishing boundary	quadratic in θ	the Fourier/Fisher metric threshold; support compatibility remains a separate row condition

Let ρ be uniform on $[-\sqrt{3}, \sqrt{3}]$, let $\mu_2 = \rho^{\otimes 2}$, and consider $T = \bigoplus_k O_{\theta_k}$. Write $\delta_k = \min_{m\in\mathbb{Z}} |\theta_k - m\pi/2| \in [0, \pi/4]$. For one block the affinity is the relative area of intersection of a square and its rotation. Removing the four corner triangles gives, for $0 < \delta \le \pi/4$,

$$\begin{aligned} \mathsf{A}(\mu_2, (O_\delta)_\#\mu_2) &= 1 - \frac{(\cos\delta + \sin\delta - 1)^2}{2\sin\delta\cos\delta}, \\ 1 - \mathsf{A}(\mu_2, (O_\delta)_\#\mu_2) &= \frac{\delta}{2} + O(\delta^2). \end{aligned} \tag{79}$$

At $\delta = 0$, the affinity is one. Each nonzero δ leaves positive area on both sides of the support difference. Consequently Theorem 6.4 gives the complete trichotomy

$$\begin{array}{c|l} \delta_k = 0 \text{ for every } k & \nu_T = \mu \\ 0 < \sum_k \delta_k < \infty & \nu_T \not\perp \mu, \quad \nu_T \not\ll \mu, \quad \mu \not\ll \nu_T \\ \sum_k \delta_k = \infty & \nu_T \perp \mu. \end{array} \tag{80}$$

In particular, $\theta_k = 1/k$ gives $T - I \in \mathcal{S}_2$ but $\nu_T \perp \mu$. With $\theta_k = 1/k^2$, the laws instead have positive affinity and neither is absolutely continuous with respect to the other.

For comparison, use the variance-one rescaling of a density proportional to $(1 - x^2)^r \mathbf{1}_{\{|x|<1\}}$, $r > 1$. Its zero-extended square root satisfies the Sobolev hypotheses. Let f be this density,

$\rho(dx) = f(x)\,dx$, $\mu_2 = \rho^{\otimes 2}$, and $g = \sqrt{f}$. Writing $J = 4\|g'\|_2^2 > 1$, the skew-generator formula gives

$$1 - \mathsf{A}(\mu_2, (O_\theta)_\# \mu_2) = \frac{J-1}{4}\theta^2 + o(\theta^2). \tag{81}$$

Indeed, the planar skew generator has Frobenius norm squared two, and $h^2 = \theta^2 \langle A, \mathcal{I}A\rangle_F/4 + o(\theta^2)$. The only rotations preserving the square support are its signed permutation symmetries. Every rotated square shares a neighborhood of the origin with the original square, on which both densities are positive; hence $\mathsf{A}(\mu_2, (O_\theta)_\#\mu_2) > 0$ for every angle. Affinity is continuous and strictly below one away from them, so the three cases in (80) now use $\sum_k \delta_k^2$ in place of $\sum_k \delta_k$. Equivalence still requires every $\delta_k = 0$. Thus the quadratic metric supplied by the Fourier/Fisher method controls accumulation of Hellinger loss, while boundary compatibility separately controls absolute continuity.

6.4 Complete linear classification for bounded and half-line laws

Write $\rho = \mathrm{Law}(X)$, $\rho^- = \mathrm{Law}(-X)$, and $\mu = \rho^{\otimes \mathbb{N}}$. Throughout this subsection X is centered with variance one. The operator action is the row-series action of Theorem 4.1.

For bounded and one-sided supports, the row condition has much stronger consequences than it first suggests. Conditional expectations of finite row sums see the boundary of the support. This turns support compatibility into an ℓ^1-type row constraint, and the inverse relations then force the operator to be monomial. The classification below therefore needs no density and no Fisher information.

Theorem 6.3 (Linear equivalence for bounded and half-line coordinate laws)**.** *Let $T \in \mathrm{GL}(H)$.*

1. *Suppose $a = \operatorname{ess\,inf} X < 0 < b = \operatorname{ess\,sup} X$ are finite. Then $\nu_T \sim \mu$ forces T to be a signed permutation. The complete equivalence group is as follows:*

relation between ρ^- and ρ	*allowed signed permutations*
$\rho^- = \rho$	*all signs*
$\rho^- \sim \rho,\ \rho^- \neq \rho$	*only finitely many negative signs*
$\rho^- \not\sim \rho$	*no negative signs.*

   In particular, $a + b \neq 0$ permits only ordinary permutations. The support need not fill an interval, and a density is not required.

2. *Suppose $\operatorname{ess\,inf} X = a > -\infty$ and $\operatorname{ess\,sup} X = +\infty$. Then*

$$\nu_T \sim \mu \quad \Longleftrightarrow \quad T \text{ is an ordinary coordinate permutation.} \tag{82}$$

   The same conclusion holds for a law bounded above and unbounded below.

In the second case every equivalence is exact invariance. Failure of any of these equivalence criteria does not, by itself, imply singularity.

Proof. Assume $\nu_T \sim \mu$, and set $S = T^{-1}$. The measurable inverse action implies $(\Phi_S)_\# \nu_T = \mu$. Pushing equivalence forward by Φ_S therefore gives $\nu_S \sim \mu$. For any row r of T or S, the random series $Z = \sum_j r_j X_j$ converges in L^2. Centering and independence imply

$$\mathbb{E}[Z \mid X_1, \dots, X_m] = \sum_{j \le m} r_j X_j. \tag{83}$$

In the bounded case $Z \in [a, b]$ almost surely. Its conditional expectation lies in the same interval. Independence and the definition of essential endpoints show that the essential width of the finite sum on the right of (83) is $(b-a)\sum_{j\le m}|r_j|$. Consequently every row of both T and S has ℓ^1-norm at most one. This proves the required bound even when the original row has infinitely many nonzero entries.

Let $A_{ij} = |T_{ij}|$ and $B_{ij} = |S_{ij}|$. The products defining the entries of TS and ST are absolutely convergent by the ℓ^2 row/column Cauchy–Schwarz inequality. Thus $AB \ge I$ and $BA \ge I$ entrywise. Tonelli's theorem and the row bounds give

$$\sum_j (AB)_{ij} = \sum_k A_{ik} \sum_j B_{kj} \le 1, \qquad \sum_j (BA)_{ij} \le 1.$$

It follows that $AB = BA = I$. This reasoning concerns nonnegative matrix entries and does not assume that entrywise absolute values define bounded operators on ℓ^2.

For completeness, nonnegative matrices inverse to one another in this sense are monomial. For each i, choose k with $A_{ik}B_{ki} > 0$. The off-diagonal entries of $AB = I$ force row k of B to vanish outside column i. Then $BA = I$ forces row i of A to vanish outside column k. The resulting correspondence is bijective. Its weights and their reciprocals are at most one by the row bounds, so all weights are one. Hence T is a signed permutation.

For a negative row sign, scalar equivalence requires $\rho^- \sim \rho$, which in particular requires $a = -b$. The remaining classification is a product calculation. Put $r = \mathsf{A}(\rho, \rho^-)$. When these laws are equivalent, $r > 0$, and $r = 1$ exactly when they are equal. Each negative sign contributes the factor r, while positive signs contribute one. Kakutani's theorem [10] gives precisely the three alternatives stated above.

For the lower-bounded, upper-unbounded case, centering and nondegeneracy give $a < 0$. Now every finite sum in (83) is bounded below by a. A negative coefficient would make its essential infimum $-\infty$, which is impossible. Thus every $r_j \ge 0$, and its finite essential infimum gives

$$a \sum_{j\le m} r_j \ge a, \qquad \text{hence} \qquad \sum_j r_j \le 1.$$

Both T and S are therefore nonnegative with row sums at most one. The same inverse-matrix argument makes T an ordinary permutation. Such a permutation preserves the i.i.d. product, proving the converse. Reflection proves the upper-bounded case. $\square$

This is the main bounded-support and half-line payoff. For $Y \sim \mathrm{Gamma}(\alpha, \theta)$, where θ is the scale, put

$$X = \frac{Y - \alpha\theta}{\sqrt{\alpha}\theta}.$$

Its essential endpoints are $-\sqrt{\alpha}$ and $+\infty$. Thus (82) gives the complete homogeneous linear equivalence classification for every $\alpha, \theta > 0$, including the shape-two Fourier example in Section 2.4: only ordinary coordinate permutations are allowed. Compact support permits signs only to the extent allowed by the scalar reflected law. For compactly supported densities positive almost everywhere inside $(-a, a)$, reflection equivalence is automatic, so an asymmetric such density permits finitely many sign changes although they do not preserve the measure exactly. A standardized asymmetric Beta law has unequal endpoint magnitudes, so it permits only permutations. These conclusions come from support geometry rather than from the Fourier/Fisher estimates.

6.5 Why the support test is indispensable

The zero-set row condition is not a cosmetic addition to the Hilbert–Schmidt criterion. It separates positive Hellinger overlap from genuine two-sided absolute continuity even for a rank-one perturbation.

Let f be the variance-one rescaling of $C(1-x^2)^2\mathbf{1}_{\{|x|<1\}}$, and write its support as $[-a,a]$. Set $\rho(dx)=f(x)\,dx$. Its zero-extended square root belongs to $W^{1,2}(\mathbb{R})$, and $x(\sqrt{f})'\in L^2(\mathbb{R})$. For $0<c<1$, set

$$T=c\oplus I,\qquad f_c(y)=c^{-1}f(y/c),\qquad \mu=\rho^{\otimes\mathbb{N}}.$$

Then $T-I$ has rank one and

$$\nu_T=(f_c(y)\,dy)\otimes\rho^{\otimes\{2,3,\dots\}}\ll\mu,\qquad \mu\not\ll\nu_T.$$

Indeed, f_c is positive on $(-ca,ca)$ and zero on the rest of $(-a,a)$. Nevertheless,

$$\mathsf{A}(\mu,\nu_T)=\int\sqrt{f(y)f_c(y)}\,dy>0,\qquad \sup_n E_n(T)<\infty.$$

Thus neither Hilbert–Schmidt proximity nor bounded projected energy implies equivalence without the inverse row condition. The failure occurs with finite location and scale Fisher information.

6.6 Exact block criteria without equivalence of the factors

For reference, the boundary examples above use the following product criterion. It separates absolute continuity within each pair of factors from accumulation of Hellinger loss, and is not part of the linear i.i.d. classification itself.

Theorem 6.4 (Block products without equivalence of the factors)**.** *Let $\mu=\bigotimes_{k\geq1}\lambda_k$ and $\nu=\bigotimes_{k\geq1}\eta_k$ be probability products over finite-dimensional coordinate blocks. Set*

$$r_k=\mathsf{A}(\lambda_k,\eta_k),\qquad L=\sum_k-\log r_k\in[0,\infty],\qquad -\log 0=+\infty.$$

Then

$$\mathsf{A}(\mu,\nu)=\prod_k r_k,\qquad \mu\perp\nu\quad\iff\quad L=\infty. \tag{84}$$

If $L<\infty$, then

$$\begin{aligned}\nu\ll\mu\quad&\iff\quad \eta_k\ll\lambda_k \text{ for every } k,\\ \mu\ll\nu\quad&\iff\quad \lambda_k\ll\eta_k \text{ for every } k.\end{aligned} \tag{85}$$

Thus positive affinity allows equivalence, either direction of strict absolute continuity, or neither direction. For a linear block $A_k\in\mathrm{GL}(d_k)$ and $\lambda_k=\rho^{\otimes d_k}$, where $\rho(dx)=f(x)\,dx$, the comparison density is

$$|\det A_k|^{-1}\prod_{i=1}^{d_k}f((A_k^{-1}y)_i).$$

The local absolute-continuity conditions concern its actual zero set, not just the closure of its support.

Proof. Affinity tensorizes on each finite block prefix. Convergence of projected affinities to global affinity follows by conditional expectation under the common dominating measure $(\mu + \nu)/2$; the proof is also given below. This gives (84).

Necessity of each local absolute-continuity condition follows by projection. For sufficiency, suppose $\eta_k \ll \lambda_k$ for all k and $\prod_k r_k > 0$. With $\ell_n = \prod_{k \le n} d\eta_k/d\lambda_k$, independence gives

$$\|\sqrt{\ell_m} - \sqrt{\ell_n}\|^2_{L^2(\mu)} = 2\left(1 - \prod_{n<k\le m} r_k\right) \longrightarrow 0.$$

The square roots have norm one, so their limit squared is a probability density. Cylinder tests identify its measure with ν. Interchanging μ, ν proves the reverse assertion. This is the product argument behind Kakutani's criterion; the mutual absolute-continuity hypothesis on each factor cannot be discarded [10, 13]. □

Corollary 6.5 (Masses of the absolutely continuous parts)**.** *Take $m_k = (\lambda_k + \eta_k)/2$, with densities p_k, q_k, and put*

$$a_k = \lambda_k(\{q_k > 0\}), \qquad b_k = \eta_k(\{p_k > 0\}).$$

If $L < \infty$, denote the absolutely continuous parts of ν relative to μ, and of μ relative to ν, by ν_a and μ_a, respectively. Then

$$\nu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k b_k, \qquad \mu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k a_k. \tag{86}$$

Proof. Since $r_k^2 \le a_k b_k$ and $L < \infty$, both displayed products are positive. On $C = \prod_k \{p_k q_k > 0\}$, the two conditional product laws have equivalent factors and affinities $r_k/\sqrt{a_k b_k}$ with positive product; hence they are equivalent. Outside C, each measure is carried by a set null for the other. Their restrictions to C are therefore exactly their absolutely continuous parts. □

Gamma scale changes and the effect of centering. For the uncentered shape-α, scale-one Gamma density, positive scaling by c preserves the scalar measure class and gives

$$\int_0^\infty \sqrt{f(y)c^{-1}f(y/c)}\,dy = \left(\frac{2\sqrt{c}}{1+c}\right)^\alpha = \cosh\left(\frac{\log c}{2}\right)^{-\alpha}. \tag{87}$$

Integration of $y^{\alpha-1}e^{-(1+c^{-1})y/2}$ proves the formula. Since $\log\cosh(u/2) \sim u^2/8$ at zero, it follows that

$$\bigotimes_k \mathrm{Law}(c_k Y) \sim \bigotimes_k \mathrm{Law}(Y) \quad \Longleftrightarrow \quad \sum_k (\log c_k)^2 < \infty, \qquad c_k > 0. \tag{88}$$

The condition is equivalently $\sum_k (c_k - 1)^2 < \infty$. This does not contradict Theorem 6.3: for $X = (Y - \alpha)/\sqrt{\alpha}$, the map $Y \mapsto cY$ becomes

$$X \longmapsto cX + \sqrt{\alpha}(c-1),$$

an affine map fixing the lower endpoint, not the homogeneous linear map $X \mapsto cX$.

Corollary 6.6 (Affine Gamma classification). *Let μ be the standardized shape-α Gamma product. For $T \in \mathrm{GL}(H)$ and $b \in \ell^2$, the law of $\Phi_T(x) + b$ under μ is equivalent to μ if and only if there are a coordinate permutation π and numbers $c_i > 0$ such that*

$$(Tx)_i = c_i x_{\pi(i)}, \qquad b_i = \sqrt{\alpha}(c_i - 1), \qquad \sum_i (\log c_i)^2 < \infty.$$

Proof. The inverse affine map has linear part T^{-1} and shift $-T^{-1}b \in \ell^2$. To justify this for the row-series versions, Cauchy–Schwarz gives $\Phi_S(z + b) = \Phi_S(z) + Sb$ wherever the selected row limits exist, for every $b \in \ell^2$. The affine orbit has second-moment bound $\|T\|_{\mathrm{op}}^2 + \|b\|_2^2$. This translation identity and the measurable composition lemma show that the inverse affine map pushes the orbit back to μ. Its pushforward of μ is consequently also equivalent to μ. Equivalence and the lower support bound show, by conditioning finite row sums as before, that both linear parts have nonnegative entries: a negative coefficient would produce an unbounded lower tail, which no finite shift can repair. Since T and T^{-1} have nonnegative matrix entries, the off-diagonal inverse identities used above show that T is a weighted permutation operator. Equality of the scalar lower endpoints forces $-\sqrt{\alpha}c_i + b_i = -\sqrt{\alpha}$. The product criterion (88) then proves necessity and sufficiency. □

6.7 An exact replacement for projected-energy equivalence

For reference, there is also a purely measure-theoretic fallback when one does not want to use the operator classification. For densities satisfying (74), Theorem 6.2 gives the sharper linear answer. The following finite-marginal criterion applies without Sobolev assumptions and without any dichotomy.

Proposition 6.7 (Two-sided finite-marginal likelihood ratio tails). *Let μ, ν be any probability measures on $\mathbb{R}^{\mathbb{N}}$, and let μ_n, ν_n be their first n marginals. Choose densities p_n, q_n with respect to any common dominating measure m_n. For $M > 1$, define*

$$D_n(M) = \int_{\{q_n > M p_n\}} q_n \, dm_n + \int_{\{p_n > M q_n\}} p_n \, dm_n.$$

Then

$$\mu \sim \nu \quad \Longleftrightarrow \quad \lim_{M\to\infty} \sup_n D_n(M) = 0. \tag{89}$$

Equivalently, every finite pair is equivalent and its forward and reverse likelihood-ratio martingales are uniformly integrable in their respective reference measures. Without any finite-marginal equivalence assumption,

$$\mathsf{A}(\mu_n, \nu_n) \downarrow \mathsf{A}(\mu, \nu), \qquad \sup_n [-2 \log \mathsf{A}(\mu_n, \nu_n)] < \infty \quad \Longleftrightarrow \quad \mu \not\perp \nu. \tag{90}$$

Proof. If the limit in (89) is zero, a set where $p_n = 0 < q_n$ must have zero ν_n-mass for every n; the reverse statement follows from the second integral. Hence $\mu_n \sim \nu_n$. Writing $\ell_n(x) = q_n(\pi_n x)/p_n(\pi_n x)$, the two terms in $D_n(M)$ are exactly

$$\mathbb{E}_\mu[\ell_n \mathbf{1}_{\{\ell_n > M\}}], \qquad \mathbb{E}_\nu[\ell_n^{-1} \mathbf{1}_{\{\ell_n^{-1} > M\}}].$$

They are the uniform-integrability tests for the two mean-one likelihood-ratio martingales. Each L^1 limit has mass one, and cylinder tests identify the corresponding absolutely continuous measure. This gives both directions of absolute continuity. Conversely, under equivalence these martingales are the conditional expectations of the full likelihood ratio and its reciprocal, so both are uniformly integrable.

For the affinity assertion put $m = (\mu + \nu)/2$, $u = d\mu/dm$, and $v = d\nu/dm$. Both densities are bounded by two. For $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, set $u_n = \mathbb{E}_m[u \mid \mathcal{F}_n]$ and $v_n = \mathbb{E}_m[v \mid \mathcal{F}_n]$. Martingale convergence and bounded convergence imply

$$\mathsf{A}(\mu_n, \nu_n) = \int \sqrt{u_n v_n}\, dm \longrightarrow \int \sqrt{uv}\, dm.$$

Conditional Cauchy–Schwarz gives monotonicity. Global affinity is zero exactly for mutually singular measures, which proves (90). $\square$

For a density f and $T \in \mathrm{GL}(H)$, each finite output marginal still has a Lebesgue density: condition on the complement of an invertible input minor as in Theorem 4.3. That density may now vanish on a different set from $\prod_{i \le n} f(x_i)$. Thus (89) is an exact criterion for arbitrary mixing even in this case. Under (74), Theorem 6.2 identifies the linear equivalence group by a Hilbert–Schmidt condition and scalar row conditions for the operator and its inverse. This includes unbounded densities with gaps. For bounded and one-sided laws, Theorem 6.3 further reduces the group to explicit classes of coordinate permutations, without Sobolev assumptions.

7 Scope and Sharpness of the Hypotheses

7.1 The finite-variance assumptions

We first return to the hypotheses of the finite-variance positive-density theorem and separate their roles. The support-aware classification of Section 6 shows that positivity is not merely a regularity convenience: without it, Hilbert–Schmidt proximity must be paired with the zero-set row condition, and boundary geometry may collapse the equivalence group to permutations. The remaining assumptions control the arbitrary-mixing Fourier/Fisher mechanism.

The finite-variance positive-density theorem is broad within its regime. It includes centered, variance-normalized Laplace densities despite their cusp at the origin, as well as variance-normalized densities proportional to e^{-x^4}. Symmetry is not needed in any step of the proof.

For examples allowing both asymmetry and heavy tails, take a density proportional to

$$(1 + y^2)^{-(\tau+1)/2} e^{\beta \arctan y}, \qquad \tau > 2, \quad \beta \in \mathbb{R}, \tag{91}$$

and center and rescale to variance one. Its location score before centering and rescaling is

$$\frac{(\tau + 1)y - \beta}{1 + y^2}.$$

Both this score and its product with y are bounded. The density is positive and smooth. It has a finite second moment. Centering and rescaling preserve finiteness of the location and scale Fisher information. So the standardized density satisfies (3). For $2 < \tau \le 4$, its fourth moment is infinite. When $\beta \neq 0$, the two tails have unequal leading constants. So centering does not make the density even. The case $\beta = 0$ is a rescaled Student t family. These examples show that neither evenness nor a fourth moment is implicit in the theorem.

The Gamma example used to illustrate the rotational Fourier defect in Section 2.4 should therefore be read through the support-aware theorem, not through the positive-density theorem.

Its centered density vanishes on a half-line. The Fourier defect still gives a local Hellinger obstruction to rotations, but equivalence is decided by the half-line support classification: homogeneous linear equivalence permits only coordinate permutations. The same support theorem classifies all bounded coordinate laws, including disconnected supports and laws without densities. Theorem 6.6 treats the distinct affine Gamma problem, where support-preserving scales do allow equivalence.

The moment assumptions play a different role from the boundary assumptions. Support controls which rows are admissible from the viewpoint of null sets; the second moment controls whether arbitrary ℓ^2 rows define the measurable linear action used throughout Theorem 1.1. It also has a specific role beyond differentiating the expectation of the product of centered complex exponentials. It makes every ℓ^2 row into a convergent random series. It supplies uniform integrability of squares of linear forms. It gives the coefficient-subsequence argument with an independent Gaussian summand in Theorem 5.5. Thus the first-moment ODE result is stronger in its moment scope than Theorem 1.1.

Returning to the finite-variance argument under (3), the Hellinger path estimates give L^1 convergence and uniform integrability of likelihood ratios. The finite real Fourier sums are bounded in each coordinate. Their dimension-free bounds are the $L^2(\mu_n)$ and generator estimates proved above. No variance bound for their aggregate under an arbitrary mixed output law is needed in the tail-symmetry argument.

7.2 The symmetric stable branch

The second-moment boundary does not leave the symmetric stable case unclassified, but it changes both the coefficient domain and the mixing cost. Fix $0 < \alpha < 2$, let ρ_α be the standard symmetric α-stable law, and put

$$\widehat{\rho}_\alpha(t) = e^{-|t|^\alpha}, \qquad \mu_\alpha = \rho_\alpha^{\otimes \mathbb{N}}, \qquad \mu_{\alpha,n} = \rho_\alpha^{\otimes n}. \tag{92}$$

Its density f_α is smooth and strictly positive, and its location and scale Fisher information are finite, although its second moment is infinite [1]. Indeed, $f_\alpha(x) \asymp (1+|x|)^{-1-\alpha}$ and its location score is $O((1+|x|)^{-1})$ in the tails, so both the location score and its product with x are square integrable against f_α. For $\alpha \le 1$, the normalization in (92) specifies a zero location parameter, not a finite mean.

A stable row series $\sum_j a_j X_j$ converges almost surely precisely when $a \in \ell^\alpha$. For a real matrix T with ℓ^α rows, write $\Phi_T^{(\alpha)}$ for its stable row-series realization and $\nu_T^{(\alpha)} = (\Phi_T^{(\alpha)})_\# \mu_\alpha$. We call this row action compatibly invertible if there is a matrix S with ℓ^α rows such that the two row-series maps are measurable and

$$\Phi_S^{(\alpha)} \circ \Phi_T^{(\alpha)} = I, \qquad \Phi_T^{(\alpha)} \circ \Phi_S^{(\alpha)} = I$$

on common conull carriers for the corresponding orbit laws. This is the stable analogue of the compatible realization supplied by Theorem 4.1; it is an assumption on the action, not a claim that every bounded ℓ^2 operator has stable row sums.

For a matrix $M = (m_{ij})$, define the anisotropic mixing cost

$$\mathfrak{e}_\alpha(M) = \sum_i |m_{ii} - 1|^2 + \sum_j \left(\sum_{i \ne j} |m_{ij}|^2 \right)^{\alpha/2}. \tag{93}$$

The first term measures coordinate scales. The second first takes the ℓ^2-norm down each off-diagonal input column and then its α-th power across source coordinates. For the actual stable output marginals, set

$$E_{\alpha,n}(T) = -2\log \mathsf{A}\big(\mu_{\alpha,n}, (\pi_n)_{\#}\nu_T^{(\alpha)}\big). \tag{94}$$

Theorem 7.1 (Symmetric stable coordinates)**.** *Let $0 < \alpha < 2$ and let T have a compatibly invertible stable row action. Then*

$$\boxed{\begin{aligned} \nu_T^{(\alpha)} \sim \mu_\alpha \quad &\iff \quad T|_H \in \mathrm{GL}(H) \text{ and } \mathfrak{c}_\alpha(Q^{-1}T) < \infty \text{ for some } Q \in \mathcal{P} \\ &\iff \quad \sup_n E_{\alpha,n}(T) < \infty, \end{aligned}} \tag{95}$$

where $\mathcal{P}$ is the full signed-permutation group and $\mathcal{P}_n$ is its finite-dimensional counterpart. If these conditions fail, then $\nu_T^{(\alpha)} \perp \mu_\alpha$ and $E_{\alpha,n}(T) \uparrow \infty$. Conversely, if $T \in \mathrm{GL}(H)$ satisfies the summability condition in (95), that condition itself supplies the stable row action and a compatible row-series inverse. The exact stabilizer in this class is $\mathcal{P}$.

Proof. We organize the proof around the four points at which the finite-variance argument must be replaced.

Dichotomy, energy, and the coefficient domain. Put $g_\alpha = \sqrt{f_\alpha}$ and define the translation affinity

$$a_\alpha(u) = \int_{\mathbb{R}} g_\alpha(x) g_\alpha(x-u)\, dx.$$

Finite positive location Fisher information, strict positivity of the density, and decay of translations in L^2 give

$$1 - a_\alpha(u) \asymp_\alpha \min\{u^2, 1\}. \tag{96}$$

Kakutani's theorem therefore identifies ℓ^2 as the exact translation-equivalence space of μ_α. If $\nu_T^{(\alpha)} \sim \mu_\alpha$, translating before and after the compatible row action shows that T and its row inverse map this translation space into itself. The closed graph theorem then gives $T|_H \in \mathrm{GL}(H)$.

The compatible two-sided inverse makes every finite family of output rows linearly independent. It therefore has a nonsingular input minor, and the remaining stable inputs contribute an independent convolution factor; hence every finite output block has a strictly positive density. The finite-marginal likelihood ratios consequently form the same mean-one martingale as in Theorem 4.3. The product measure is quasi-invariant and ergodic under c_{00}, while its row-series image has the same properties under $T(c_{00})$. The equivalent–singular theorem for quasi-invariant ergodic measures allows these two translation spaces to differ and therefore applies [11]. It follows exactly as in (43) that

$$\nu_T^{(\alpha)} \sim \mu_\alpha \quad \iff \quad \sup_n E_{\alpha,n}(T) < \infty, \tag{97}$$

with singularity and divergent energy on the other branch.

Sufficiency. For a matrix K, put

$$F_\alpha(K) = \sum_j \|Ke_j\|_2^\alpha, \qquad \mathcal{C}_\alpha = \{K : F_\alpha(K) < \infty\}.$$

This column class satisfies

$$F_\alpha(AK) \le \|A\|_{\mathrm{op}}^\alpha F_\alpha(K), \qquad F_\alpha(KD) \le \|D\|_{\mathrm{op}}^\alpha F_\alpha(K) \tag{98}$$

for bounded A and bounded diagonal D, and

$$\|K^*b\|_\alpha^\alpha \le F_\alpha(K)\|b\|_2^\alpha. \tag{99}$$

Thus $K \in \mathcal{C}_\alpha$ has admissible stable rows.

The finite-dimensional estimate that replaces the matrix Fisher upper bound is

$$h^2(A_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \le C_\alpha \min\{\mathfrak{e}_\alpha(A), 1\} \tag{100}$$

for every invertible real $n \times n$ matrix A, with a constant independent of n. To prove it, put $\beta = \alpha/2$, let V_i be independent positive β-stable variables with $\mathbb{E}e^{-tV_i} = e^{-t^\beta}$, and write $X_i = \sqrt{2V_i}Z_i$ with independent standard Gaussians Z_i. Conditionally on $V = \operatorname{diag}(V_i)$, Hellinger data processing and the Gaussian affinity formula bound the left side of (100) by a constant times

$$\mathbb{E}\min\{\|V^{-1/2}(A-I)V^{1/2}\|_F^2, 1\}.$$

For $c_j^2 = \sum_{i \ne j} |a_{ij}|^2$, the contribution of source column j is

$$W_j = V_j \sum_{i \ne j} \frac{|a_{ij}|^2}{V_i}.$$

The positive stable estimates

$$\mathbb{P}(V > r) \le C_\beta r^{-\beta}, \qquad \mathbb{E}[V; V \le r] \le C_\beta r^{1-\beta}, \qquad \mathbb{E}V^{-1} < \infty$$

give

$$\mathbb{P}(c_j^2 V_j > 1) + \mathbb{E}[W_j; c_j^2 V_j \le 1] \le C_\alpha c_j^\alpha.$$

Summing over source columns, while retaining the ordinary quadratic bound on the diagonal, proves (100).

Now remove the measure-preserving Q and write the resulting matrix as $M = D + K$, where $D = \operatorname{diag}(d_i)$, $(d_i - 1) \in \ell^2$, and $K \in \mathcal{C}_\alpha$. The finitely many indices on which the diagonal of M is not positive are absorbed into K. For

$$D_n = I + P_n(D-I)P_n, \qquad K_n = P_n K P_n, \qquad M_n = D_n + K_n,$$

we have $M_n \to M$ in operator norm, so the M_n are eventually invertible with uniformly bounded inverses. Write

$$M_n^{-1} = D_n^{-1} + J_n, \qquad J_n = -M_n^{-1}K_n D_n^{-1}.$$

Then $F_\alpha(J_n)$ is uniformly bounded, and

$$\begin{aligned} M_n^{-1}M_m - I = D_n^{-1}(D_m - D_n) + J_n(D_m - D_n) \\ + M_n^{-1}(K_m - K_n). \end{aligned} \tag{101}$$

The first term has diagonal ℓ^2-norm tending to zero. The other two tend to zero in $\mathcal{C}_\alpha$ by (98), $\|D_m - D_n\|_{\mathrm{op}} \to 0$, and $F_\alpha(K_m - K_n) \to 0$; their diagonal square cost tends to zero as well because $\mathcal{C}_\alpha \subset \mathcal{S}_2$. Thus (100), applied to $M_n^{-1}M_m$, makes the square-root likelihoods of $(M_n)_\# \mu_\alpha$ Cauchy in $L^2(\mu_\alpha)$. Their limit has norm one, and convergence of each stable row in

ℓ^α identifies its square with $d\nu_M^{(\alpha)}/d\mu_\alpha$. This proves absolute continuity and then equivalence by (97). Finally,

$$(D+K)^{-1} = D^{-1} - (D+K)^{-1}KD^{-1} \tag{102}$$

and (98) show that the inverse has the same admissible form. The continuity of the stable random functional from ℓ^α to convergence in probability lets the finite-matrix composition identities pass to the row-series limits, producing compatible measurable versions. This also proves the last assertion of the theorem.

The global signed-permutation match. Suppose conversely that $\nu_T^{(\alpha)} \sim \mu_\alpha$. We transfer the problem to a finite-variance product without applying the finite-variance theorem directly to the stable law. Let $T^{-*} := (T^{-1})^*$. Let G_α be the unitary Fourier transform of g_α, set

$$\lambda(d\xi) = |G_\alpha(\xi)|^2\, d\xi, \qquad \Lambda_k = (\lambda^{*k})^{\otimes\mathbb{N}}.$$

The probability λ is symmetric and has variance $\|g'_\alpha\|_2^2 \in (0,\infty)$. Moreover, $G_\alpha * G_\alpha$ is a positive constant multiple of $e^{-|t|^\alpha}$; hence $|G_\alpha|^2 * |G_\alpha|^2 > 0$ almost everywhere. The characteristic function of λ is the translation affinity a_α. Polynomial stable tails give, for some $p, \delta > 0$,

$$c(1+|t|)^{-p} \le a_\alpha(t) \le C(1+|t|)^{-\delta},$$

and a'_α is bounded. Choose an integer r so large that

$$\int_{\mathbb{R}} t^2\big(|a_\alpha(t)|^{2r} + |(a_\alpha^r)'(t)|^2\big)\, dt < \infty.$$

Fourier inversion makes the density of λ^{*r} a bounded-variation density; three such convolutions have finite location Fisher information by the Fisher smoothing estimate [1]. Let b be the density of λ^{*3r}, put $q = b * b$, and let s_b, s_q be their location scores. Independent U, V with density b satisfy

$$xs_q(x) - 1 = 2\mathbb{E}[Us_b(V) \mid U+V = x].$$

Their finite variance and the finite Fisher information of b therefore give finite scale Fisher information for q. Thus for the even integer $k = 6r$, the standardized density of λ^{*k} is positive and satisfies (3). It is non-Gaussian because its characteristic function a_α^k has a polynomial lower bound, whereas a nondegenerate Gaussian characteristic function has Gaussian decay.

Let $p_n = f_\alpha^{\otimes n}$, let $q_{T,n}$ be the density of the true first n stable outputs, and put

$$L_n = \frac{q_{T,n}}{p_n}, \qquad \psi_n = \sqrt{q_{T,n}}, \qquad \Omega_n = g_\alpha^{\otimes n}.$$

Here L_n is lifted to the product space and $\psi_n = \Omega_n\sqrt{L_n}$. Equivalence gives $r_n := \sqrt{L_n} \to r := \sqrt{L}$ in $L^2(\mu_\alpha)$.

For each n, take k tensor copies of ψ_n, apply the finite-dimensional unitary Fourier transform, square the modulus, and push the resulting probability on $\mathbb{R}^{kn}$ forward by adding the k frequency coordinates belonging to each site. Call the resulting probability on $\mathbb{R}^n$ η_n, and extend it to the full sequence space by

$$\sigma_n = \eta_n \otimes (\lambda^{*k})^{\otimes\{i>n\}}. \tag{103}$$

The finite-dimensional pushforward η_n has a Lebesgue density, and the density of λ^{*k} is positive almost everywhere; hence $\sigma_n \ll \Lambda_k$.

For $m > n$, pad ψ_n by $m-n$ factors of g_α before making the Fourier comparison. Write $\|\cdot\|_{\mathrm{TV},1}$ for total variation in the L^1-norm convention. Unitarity, the inequality $\||u|^2 - |v|^2\|_1 \le 2\|u-v\|_2$ for unit vectors, and contraction of total variation under pushforward give

$$\|\sigma_m - \sigma_n\|_{\mathrm{TV},1} \le 2\|r_m^{\otimes k} - r_n^{\otimes k}\|_{L^2(\mu_\alpha^{\otimes k})} \le 2k\|r_m - r_n\|_{L^2(\mu_\alpha)}. \tag{104}$$

Thus σ_n converges in total variation to a probability $\sigma \ll \Lambda_k$.

Let $\nu^{(n)} = L_n\mu_\alpha$; this measure has the true first n output marginal and the original independent tail. For every finitely supported h, Fourier autocorrelation and $r_n \to r$ give

$$\begin{aligned}\widehat{\sigma}(h) &= \lim_n \mathsf{A}\big(\nu^{(n)}, (x \mapsto x+h)_\#\nu^{(n)}\big)^k \\ &= \mathsf{A}\big(\nu_T^{(\alpha)}, (x \mapsto x+h)_\#\nu_T^{(\alpha)}\big)^k \\ &= \prod_i a_\alpha((T^{-1}h)_i)^k.\end{aligned}$$

The last expression is the characteristic function of $(T^{-*})_\#\Lambda_k$, so finite-dimensional uniqueness identifies $\sigma = (T^{-*})_\#\Lambda_k$. Consequently

$$(T^{-*})_\#\Lambda_k \ll \Lambda_k. \tag{105}$$

The equivalent–singular alternative in Theorem 1.1 upgrades this absolute continuity to equivalence. Applying that theorem to the standardized λ^{*k} product then gives a signed permutation Q such that $T^{-*} - Q \in \mathcal{S}_2$. Since

$$T - Q = -T(T^{-*} - Q)^* Q,$$

we obtain $T - Q \in \mathcal{S}_2$. After removing Q, write

$$Q^{-1}T = I + H, \qquad H \in \mathcal{S}_2. \tag{106}$$

The fractional column necessity. It remains to strengthen (106) from a quadratic conclusion to the second sum in (93). The local input is the dimension-free rare-source estimate valid for every dimension and every real square matrix B: there are $\varepsilon_\alpha, c_\alpha > 0$ such that

$$\operatorname{diag} B = 0, \quad F_\alpha(B) \le \varepsilon_\alpha \quad \Longrightarrow \quad h^2((I+B)_\#\mu_{\alpha,n}, \mu_{\alpha,n}) \ge c_\alpha F_\alpha(B). \tag{107}$$

Here is the rare-source construction. Write $F = F_\alpha(B)$, $c_j = \|Be_j\|_2$, and $u_j = Be_j/c_j$ when $c_j \ne 0$. Then $\|u_j\|_2 = 1$, $(u_j)_j = 0$, and the stable scale contributed to output row i satisfies

$$r_i := \sum_j |b_{ij}|^\alpha \le F. \tag{108}$$

First, for any finite family of vectors v_j in a real Hilbert space, the stable random sum $S = \sum_j v_j X_j$ obeys

$$\mathbb{P}(\|S\|_2 > t) \le C_\alpha t^{-\alpha} \sum_j \|v_j\|_2^\alpha. \tag{109}$$

Indeed, exclude the events $\|v_j\|_2|X_j| > t$, truncate on their complement, and apply Chebyshev's inequality to the centered truncated sum. The stable tail and truncated second moment give (109). In particular, there is a deterministic $\eta_\alpha(s) \downarrow 0$ as $s \downarrow 0$ such that

$$\mathbb{E}\min\left\{\left\|\sum_j v_j X_j\right\|_2, 1\right\} \le \eta_\alpha\left(\sum_j \|v_j\|_2^\alpha\right). \tag{110}$$

Second, one bounded statistic detects a unit displacement in every Euclidean direction, uniformly in the dimension. Take $\psi(x) = \tanh x$. There are a smooth odd increasing function χ, bounded in modulus by one, and $\kappa > 0$, depending only on ρ_α, such that for every unit vector u,

$$Z_u(x) = \sum_i u_i \psi(x_i), \qquad R_u(t) = \mathbb{E}\chi(Z_u(X + tu)), \qquad R_u(t) \geq \kappa \quad (1 \leq t \leq 2). \tag{111}$$

To verify this, note that $Z_u(X)$ has mean zero and a variance independent of u. Choose a common interval carrying most of its mass and then choose χ whose derivative has a positive lower bound on the enlarged interval. Since

$$\inf_{|z| \leq 2} \mathbb{E}\psi'(X + z) > 0,$$

differentiation gives a dimension-free positive lower bound for

$$R_u'(t) = \mathbb{E}\left[\chi'(Z_u(X + tu)) \sum_i u_i^2 \psi'(X_i + tu_i)\right]$$

when $|t| \leq 2$, after discarding the common small exceptional event. Oddness then proves (111).

Third, choose a nonzero smooth odd function w, supported where $1 \leq |t| \leq 2$, with $0 \leq w \leq 1$ on the positive half-line. Put

$$Z_j(x) = \sum_{i \neq j} (u_j)_i \psi(x_i), \qquad [z]_{[-1,1]} = \max\{-1, \min\{z, 1\}\},$$

and define the bounded test

$$\Phi_B(x) = \left[\sum_{j : c_j \neq 0} w(c_j x_j) \chi(Z_j(x))\right]_{[-1,1]}. \tag{112}$$

For all sufficiently small c_j, stable tail asymptotics give

$$\mathbb{P}(1 \leq |c_j X_j| \leq 2) \leq C_\alpha c_j^\alpha, \qquad \mathbb{E}|w(c_j X_j)| \geq c_\alpha c_j^\alpha. \tag{113}$$

The gates depend on independent input coordinates. If K is the number of open gates, then $\mathbb{E}K(K-1) \leq C_\alpha F^2$. Before the outer truncation, each summand in (112) has mean zero because its gate variable is independent of its receiver variables. The truncation changes the sum only when at least two gates are open, so

$$|\mathbb{E}\Phi_B(X)| \leq C_\alpha F^2. \tag{114}$$

It remains to control the same statistic at $Y = (I + B)X$. For fixed $0 < \delta < 1/4$, (108) and the stable tail bound imply

$$\mathbb{P}\{\exists i : \ |c_i(BX)_i| > \delta\} \leq C_\alpha \delta^{-\alpha} \sum_i c_i^\alpha r_i \leq C_\alpha \delta^{-\alpha} F^2. \tag{115}$$

On the complementary event, Lipschitz continuity of w and (113) show that replacing every output gate $w(c_j Y_j)$ by its input gate $w(c_j X_j)$ changes the expectation by at most

$$C_\alpha \delta F + C_\alpha \delta^{-\alpha} F^2. \tag{116}$$

After this replacement, removing the outer truncation costs at most another $C_\alpha F^2$.

Finally, set $B^{(j)} = B(I - e_j \otimes e_j)$. The random vector $B^{(j)}X$ is independent of the gate variable X_j, and (110) shows that replacing $Z_j(Y)$ by $Z_j(X + c_j X_j u_j)$ costs, after summing over j, at most

$$C_\alpha F \eta_\alpha(F). \tag{117}$$

Because $(u_j)_j = 0$, conditioning on X_j and using (**??**) gives

$$\sum_j \mathbb{E}[w(c_j X_j)\chi(Z_j(X + c_j X_j u_j))] \geq c_\alpha F. \tag{118}$$

Choose δ and then F sufficiently small. Equations (**??**) give

$$\mathbb{E}\Phi_B(Y) - \mathbb{E}\Phi_B(X) \geq c_\alpha F.$$

The test can be nonzero only when a gate is open. The coordinates of Y have stable scales bounded by $(1+F)^{1/\alpha}$, so

$$\mathbb{E}\Phi_B(X)^2 + \mathbb{E}\Phi_B(Y)^2 \leq C_\alpha F.$$

The Hellinger Cauchy–Schwarz inequality now gives $h^2((I+B)_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F$, which is (107).

We use this estimate through tail sign symmetries, which avoids any triangularity assumption. Let B have ℓ^α rows, zero diagonal, $\|B\|_{\mathrm{op}} \leq 1/10$, and belong to $\mathcal{S}_2$. Suppose the row action of $I + B$ and its row-series inverse are compatible and $(I+B)_\# \mu_\alpha \sim \mu_\alpha$. For a finite coordinate set J, let S_J change the signs in J and set

$$\Delta_J = S_J B S_J - B, \qquad C_J = (I+B)^{-1} S_J (I+B) S_J.$$

The matrix Δ_J contains precisely the entries crossing the cut between J and its complement. Because J is finite, these entries form a sum of finitely many ℓ^2 columns and finitely many ℓ^α rows; hence $\Delta_J \in \mathcal{C}_\alpha$. The finite-dimensional bound (107) extends to such infinite matrices by applying it to $P_n \Delta_J P_n$ and using the Hellinger convergence from the already proved sufficiency argument.

Put

$$K_J = (I+B)^{-1}\Delta_J, \qquad D_J = I + \operatorname{diag} K_J, \qquad \widetilde{B}_J = D_J^{-1}(I + K_J) - I.$$

Then $C_J = I + K_J$ and $\operatorname{diag} \widetilde{B}_J = 0$. Since $\|(I+B)^{-1} - I\|_{\mathrm{op}} \leq 1/9$, orthogonal projection away from the diagonal entry in each column gives

$$\frac{8}{9}\|\Delta_J e_j\|_2 \leq \|(I - e_j \otimes e_j) K_J e_j\|_2 \leq \frac{10}{9}\|\Delta_J e_j\|_2.$$

For small $F_\alpha(\Delta_J)$, the diagonal entries of D_J lie in $[1/2, 3/2]$, and consequently

$$F_\alpha(\widetilde{B}_J) \asymp_\alpha F_\alpha(\Delta_J), \qquad \|\operatorname{diag} K_J\|_2 \leq \frac{1}{9} F_\alpha(\Delta_J)^{1/\alpha}. \tag{119}$$

The rare-source lower bound for $\widetilde{B}_J$, the diagonal scale upper bound for D_J, and the Hellinger triangle inequality now give

$$F_\alpha(\Delta_J) \leq \varepsilon_* \quad \Longrightarrow \quad h((C_J)_\# \mu_\alpha, \mu_\alpha) \geq c_* F_\alpha(\Delta_J)^{1/2}. \tag{120}$$

Indeed, the two terms have respective orders $F_\alpha(\Delta_J)^{1/2}$ and $F_\alpha(\Delta_J)^{1/\alpha}$, and the latter is smaller for small cost because $\alpha < 2$.

On the other hand, finite-prefix approximation of the likelihood ratio gives the uniform tail symmetry

$$\sup_{\substack{J\subset\{N+1,N+2,\dots\}\\ |J|<\infty}} h\big((I+B)_\#\mu_\alpha, (S_J)_\#(I+B)_\#\mu_\alpha\big) \longrightarrow 0. \tag{121}$$

Hellinger invariance and $(S_J)_\#\mu_\alpha = \mu_\alpha$ also give the connecting identity

$$h\big((I+B)_\#\mu_\alpha, (S_J)_\#(I+B)_\#\mu_\alpha\big) = h(\mu_\alpha, (C_J)_\#\mu_\alpha). \tag{122}$$

If $F_\alpha(B) = \infty$, deleting finitely many columns changes the cost by a finite amount, and so does deleting finitely many rows because those rows lie in ℓ^α. Hence every tail still has infinite column cost. Let E be a finite coordinate set in such a tail, and form $J \subset E$ by independently including each coordinate with probability one half. Column by column, concavity of $t^{\alpha/2}$ gives

$$\mathbb{E}_J F_\alpha(P_E\Delta_J P_E) \geq 2^{\alpha-1} F_\alpha(P_E B P_E). \tag{123}$$

As E increases through finite subsets of the tail, the right side is unbounded, so some finite cuts have arbitrarily large cost. Equivalence also makes every sufficiently remote one-coordinate output marginal close to ρ_α, so

$$r_i := \sum_{j\neq i} |b_{ij}|^\alpha \longrightarrow 0.$$

Together with $B \in \mathcal{S}_2$, which gives $c_i := \|Be_i\|_2 \to 0$, this yields

$$F_\alpha(\Delta_{\{i\}}) = 2^\alpha(c_i^\alpha + r_i) \longrightarrow 0.$$

Set $\gamma = \max\{1, \alpha\}$ and $\mathcal{N}(A) = F_\alpha(A)^{1/\gamma}$. For $\alpha \leq 1$, subadditivity of the α-power makes $\mathcal{N}$ subadditive; for $\alpha \geq 1$, it is the $\ell^\alpha(\ell^2)$ norm and satisfies the triangle inequality. Moreover, when $i \notin J$,

$$\Delta_{J\cup\{i\}} - \Delta_J = S_J \Delta_{\{i\}} S_J,$$

and sign multiplication preserves $\mathcal{N}$. Thus the $\mathcal{N}$-increment from adding one sufficiently remote coordinate tends uniformly to zero. Fix $0 < \eta < \varepsilon_*/2$. Starting with the empty cut and adding the elements of a finite cut supplied by (123), stop the first time the corresponding $\mathcal{N}$-threshold is crossed. For every sufficiently remote tail this produces a finite J with $\eta \leq F_\alpha(\Delta_J) \leq 2\eta$. Equations (**??**) then contradict one another. Hence $F_\alpha(B) < \infty$.

Finally, put $R = I + H$ in (106) and let $R_n = I + P_n H P_n$. For a sufficiently large n, the equivalent transformation $A = R_n^{-1}R$ is arbitrarily close to the identity in operator norm. Indeed, R_n changes only finitely many coordinates, so $(R_n)_\#\mu_\alpha \sim \mu_\alpha$; compatible composition with $R_\#\mu_\alpha \sim \mu_\alpha$ gives $A_\#\mu_\alpha \sim \mu_\alpha$. Its diagonal $D = \operatorname{diag}(a_{ii})$ is positive, boundedly invertible, and $D - I \in \mathcal{S}_2$; the one-coordinate scale criterion therefore gives $D_\#\mu_\alpha \sim \mu_\alpha$. Consequently

$$D^{-1}A = I + B, \qquad \operatorname{diag} B = 0, \quad B \in \mathcal{S}_2, \quad \|B\|_{\mathrm{op}} \leq 1/10, \quad (I+B)_\#\mu_\alpha \sim \mu_\alpha.$$

Finite-coordinate and bounded diagonal composition preserve the ℓ^α row condition and the compatible row action, so this normalized B also satisfies the hypotheses of the sign-cut argument. The sign-cut argument gives $B \in \mathcal{C}_\alpha$. Writing $R_n = I + F_n$, where F_n is supported on a finite coordinate block, we have

$$R = R_n D(I+B) = D + \big(F_n D + DB + F_n DB\big).$$

The term in parentheses belongs to $\mathcal{C}_\alpha$ by (98) and finite-coordinate support. Restoring the initial signed permutation therefore gives

$$\sum_j \left(\sum_{i \neq j} |h_{ij}|^2 \right)^{\alpha/2} < \infty,$$

while $H \in \mathcal{S}_2$ already gives $\sum_i |h_{ii}|^2 < \infty$. This is precisely $\mathfrak{e}_\alpha(Q^{-1}T) < \infty$, and completes the necessity.

If $\nu_T^{(\alpha)} = \mu_\alpha$, comparison of the joint characteristic functions gives $\|T^* u\|_\alpha = \|u\|_\alpha$ for every finitely supported u. The same statement for the compatible inverse makes T^* a surjective isometry of the stable coefficient space. For $0 < \alpha < 2$, the equality case of

$$2\|u\|_\alpha^\alpha + 2\|v\|_\alpha^\alpha - \|u+v\|_\alpha^\alpha - \|u-v\|_\alpha^\alpha \geq 0$$

shows that disjointly supported vectors have disjointly supported images. The images of the coordinate vectors therefore have pairwise disjoint supports; surjectivity makes each support a singleton, and norm preservation makes its nonzero entry ± 1. Thus $T \in \mathcal{P}$, while every member of $\mathcal{P}$ plainly preserves μ_α. This proves the stabilizer assertion and completes the proof. $\square$

7.3 Anisotropic Hellinger geometry and sharpness

The two summands in (93) have genuinely different homogeneities. For a finite matrix A, put

$$\mathcal{E}_{\alpha,n}(A) = \min_{Q \in \mathcal{P}_n} \mathfrak{e}_\alpha(Q^{-1}A).$$

If all singular values of A lie in a fixed interval $[a,b] \subset (0,\infty)$, then constants depending only on α, a, b, and not on the dimension, satisfy

$$c_{\alpha,a,b} \min\{\mathcal{E}_{\alpha,n}(A), 1\} \leq h^2(A_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_{\alpha,a,b} \min\{\mathcal{E}_{\alpha,n}(A), 1\}. \tag{124}$$

The upper bound is (100), after removing the minimizing signed permutation. For the even convolution power k constructed in the proof of Theorem 7.1, the finite-dimensional form of the Fourier transfer is the quantitative inequality

$$h^2((A^{-*})_\# (\lambda^{*k})^{\otimes n}, (\lambda^{*k})^{\otimes n}) \leq k\, h^2(A_\# \mu_{\alpha,n}, \mu_{\alpha,n}). \tag{125}$$

where $A^{-*} = (A^{-1})^*$. Indeed, the unitary Fourier transform intertwines the half-density actions of A and A^{-*}; taking moduli is an L^2 contraction. For k tensor copies, squared Hellinger distance grows by at most the factor k, and coordinatewise frequency addition is a data-processing map. Apply the finite-variance coercivity theorem to the standardized λ^{*k} product. When the singular values of A lie in $[a,b]$, those of A^{-*} lie in $[b^{-1}, a^{-1}]$, and

$$A - Q = -A(A^{-*} - Q)^* Q$$

shows that distance from the signed-permutation group changes by at most a factor depending on a, b. Thus (125) gives

$$h^2(A_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_{\alpha,a,b} \min\left\{ \min_{Q \in \mathcal{P}_n} \|A - Q\|_F^2, 1 \right\}. \tag{126}$$

Near a signed permutation, remove that permutation and write

$$d = \sum_i |a_{ii} - 1|^2, \qquad F = \sum_j \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{\alpha/2}.$$

For $d+F$ small, the identity is the nearest signed permutation, so (126) controls d. If $D = \operatorname{diag}(a_{ii})$ and $D^{-1}A = I + B$, then $\operatorname{diag} B = 0$ and $F_\alpha(B) \asymp_\alpha F$. The rare-source bound and the diagonal scale upper bound give

$$h(A_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha \sqrt{F} - C_\alpha \sqrt{d}.$$

If d dominates F, use (126); otherwise absorb the second term in the last display. This proves the local lower bound in (124). If uniform separation failed away from that neighborhood, one could take a block-diagonal sum of counterexamples whose squared Hellinger distances are summable. Kakutani's theorem would make the resulting infinite stable products equivalent. The Hilbert–Schmidt part of Theorem 7.1 forces its global matching permutation to preserve every sufficiently late block; otherwise each crossed block contributes a fixed positive square error. The restrictions of that permutation to the late blocks would then make their anisotropic costs summable, a contradiction. This proves the global lower bound.

The different local powers are already visible in two dimensions. For the shear $T_t(x, y) = (x + ty, y)$, conditioning on y and applying Plancherel gives the exact formula

$$h^2((T_t)_\# \mu_{\alpha,2}, \mu_{\alpha,2}) = 2 \int_{\mathbb{R}} (1 - e^{-|t\xi|^\alpha}) |G_\alpha(\xi)|^2 \, d\xi. \tag{127}$$

Consequently

$$h^2((T_t)_\# \mu_{\alpha,2}, \mu_{\alpha,2}) \sim 2|t|^\alpha \int_{\mathbb{R}} |\xi|^\alpha |G_\alpha(\xi)|^2 \, d\xi, \qquad t \to 0,$$

whereas a diagonal scale perturbation has squared Hellinger cost of order t^2. Thus the stable geometry is not generated by one finite matrix Fisher quadratic form.

In particular, Hilbert–Schmidt proximity is necessary but no longer sufficient. On independent two-coordinate blocks, let

$$T_a(x_k, y_k)_{k \geq 1} = (x_k + a_k y_k, y_k)_{k \geq 1}.$$

Then $T_a - I \in \mathcal{S}_2$ exactly when $a \in \ell^2$, while Theorem 7.1 gives $(T_a)_\# \mu_\alpha \sim \mu_\alpha$ exactly when $a \in \ell^\alpha$. Any $a \in \ell^2 \setminus \ell^\alpha$ therefore gives a Hilbert–Schmidt perturbation with singular image law, even though the stable coordinate density is everywhere positive and has finite location and scale Fisher information. Conversely, when one source is mixed into many outputs, the criterion charges the ℓ^2-norm of that whole column only once. An entrywise ℓ^α condition would therefore be too strong. The correct stable criterion is an anisotropic, coordinate-sensitive cost, not a Schatten-class replacement for the finite-variance theorem.

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